Pipeline

March 27, 2025

1 Generación de datos sintética con Transformers - Huggingface

Este notebook realiza un análisis exploratorio de los datos de tráfico y emisiones contaminantes, seguido de la creación de un pipeline utilizando Hugging Face.

1.0.1 Objetivos

- [x] Realizar un EDA (Análisis Exploratorio de Datos).
- [x] Preprocesar los datos y realizar imputación de valores nulos.
- [ ] Crear un pipeline con Hugging Face para predicción o clasificación.
- [ ] Explorar el TabTransformer

[52]:
```python
# Importar bibliotecas necesarias
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from transformers import pipeline
```

[53]:
```python
# Cargar los datos de los archivos CSV
trafico_feb22_ago23 = pd.read_csv('trafico_feb22_ago23.csv')
from pprint import pprint
pprint(list(trafico_feb22_ago23.columns))
```

```
['date',
 'vehicles_CAP_OUT',
 'vehicles_CAP_OUT_Zona_Granada',
 'vehicles_CAP_OUT_Zona_Catalunia_y_Otras',
 'vehicles_CAP_OUT_Zona_Andalucia_no_GR',
 'vehicles_CAP_OUT_Extranjero',
 'vehicles_CAP_OUT_Zona_Comunidad_de_Madrid',
 'vehicles_CAP_OUT_Zona_Extremadura_y_Otras',
 'vehicles_CAP_OUT_Zona_Otras',
 'vehicles_CAP_OUT_5_seats',
 'vehicles_CAP_OUT_+5_seats',
 'vehicles_CAP_OUT_-5_seats',
 'vehicles_CAP_OUT_nan_seats',
```

```
'vehicles_CAP_OUT_101-200_CO2',
'vehicles_CAP_OUT_0-100_CO2',
'vehicles_CAP_OUT_nan_CO2',
'vehicles_CAP_OUT_201-300_CO2',
'vehicles_CAP_OUT_+300_CO2',
'vehicles_CAP_IN',
'vehicles_CAP_IN_Zona_Granada',
'vehicles_CAP_IN_Zona_Catalunia_y_Otras',
'vehicles_CAP_IN_Zona_Andalucia_no_GR',
'vehicles_CAP_IN_Extranjero',
'vehicles_CAP_IN_Zona_Comunidad_de_Madrid',
'vehicles_CAP_IN_Zona_Extremadura_y_Otras',
'vehicles_CAP_IN_Zona_Otras',
'vehicles_CAP_IN_5_seats',
'vehicles_CAP_IN_+5_seats',
'vehicles_CAP_IN_-5_seats',
'vehicles_CAP_IN_nan_seats',
'vehicles_CAP_IN_101-200_CO2',
'vehicles_CAP_IN_0-100_CO2',
'vehicles_CAP_IN_nan_CO2',
'vehicles_CAP_IN_201-300_CO2',
'vehicles_CAP_IN_+300_CO2',
'vehicles_PAM_1_OUT',
'vehicles_PAM_1_OUT_Zona_Granada',
'vehicles_PAM_1_OUT_Zona_Catalunia_y_Otras',
'vehicles_PAM_1_OUT_Zona_Andalucia_no_GR',
'vehicles_PAM_1_OUT_Extranjero',
'vehicles_PAM_1_OUT_Zona_Comunidad_de_Madrid',
'vehicles_PAM_1_OUT_Zona_Extremadura_y_Otras',
'vehicles_PAM_1_OUT_Zona_Otras',
'vehicles_PAM_1_OUT_5_seats',
'vehicles_PAM_1_OUT_+5_seats',
'vehicles_PAM_1_OUT_-5_seats',
'vehicles_PAM_1_OUT_nan_seats',
'vehicles_PAM_1_OUT_101-200_CO2',
'vehicles_PAM_1_OUT_0-100_CO2',
'vehicles_PAM_1_OUT_nan_CO2',
'vehicles_PAM_1_OUT_201-300_CO2',
'vehicles_PAM_1_OUT_+300_CO2',
'vehicles_PAM_1_IN',
'vehicles_PAM_1_IN_Zona_Granada',
'vehicles_PAM_1_IN_Zona_Catalunia_y_Otras',
'vehicles_PAM_1_IN_Zona_Andalucia_no_GR',
'vehicles_PAM_1_IN_Extranjero',
'vehicles_PAM_1_IN_Zona_Comunidad_de_Madrid',
'vehicles_PAM_1_IN_Zona_Extremadura_y_Otras',
'vehicles_PAM_1_IN_Zona_Otras',
'vehicles_PAM_1_IN_5_seats',
```

```
'vehicles_PAM_1_IN_+5_seats',
'vehicles_PAM_1_IN_-5_seats',
'vehicles_PAM_1_IN_nan_seats',
'vehicles_PAM_1_IN_101-200_CO2',
'vehicles_PAM_1_IN_0-100_CO2',
'vehicles_PAM_1_IN_nan_CO2',
'vehicles_PAM_1_IN_201-300_CO2',
'vehicles_PAM_1_IN_+300_CO2',
'vehicles_PAM_2_OUT',
'vehicles_PAM_2_OUT_Zona_Granada',
'vehicles_PAM_2_OUT_Zona_Catalunia_y_Otras',
'vehicles_PAM_2_OUT_Zona_Andalucia_no_GR',
'vehicles_PAM_2_OUT_Extranjero',
'vehicles_PAM_2_OUT_Zona_Comunidad_de_Madrid',
'vehicles_PAM_2_OUT_Zona_Extremadura_y_Otras',
'vehicles_PAM_2_OUT_Zona_Otras',
'vehicles_PAM_2_OUT_5_seats',
'vehicles_PAM_2_OUT_+5_seats',
'vehicles_PAM_2_OUT_-5_seats',
'vehicles_PAM_2_OUT_nan_seats',
'vehicles_PAM_2_OUT_101-200_CO2',
'vehicles_PAM_2_OUT_0-100_CO2',
'vehicles_PAM_2_OUT_nan_CO2',
'vehicles_PAM_2_OUT_201-300_CO2',
'vehicles_PAM_2_OUT_+300_CO2',
'vehicles_PAM_2_IN',
'vehicles_PAM_2_IN_Zona_Granada',
'vehicles_PAM_2_IN_Zona_Catalunia_y_Otras',
'vehicles_PAM_2_IN_Zona_Andalucia_no_GR',
'vehicles_PAM_2_IN_Extranjero',
'vehicles_PAM_2_IN_Zona_Comunidad_de_Madrid',
'vehicles_PAM_2_IN_Zona_Extremadura_y_Otras',
'vehicles_PAM_2_IN_Zona_Otras',
'vehicles_PAM_2_IN_5_seats',
'vehicles_PAM_2_IN_+5_seats',
'vehicles_PAM_2_IN_-5_seats',
'vehicles_PAM_2_IN_nan_seats',
'vehicles_PAM_2_IN_101-200_CO2',
'vehicles_PAM_2_IN_0-100_CO2',
'vehicles_PAM_2_IN_nan_CO2',
'vehicles_PAM_2_IN_201-300_CO2',
'vehicles_PAM_2_IN_+300_CO2',
'vehicles_BUB_OUT',
'vehicles_BUB_OUT_Zona_Granada',
'vehicles_BUB_OUT_Zona_Catalunia_y_Otras',
'vehicles_BUB_OUT_Zona_Andalucia_no_GR',
'vehicles_BUB_OUT_Extranjero',
'vehicles_BUB_OUT_Zona_Comunidad_de_Madrid',
```

```
'vehicles_BUB_OUT_Zona_Extremadura_y_Otras',
'vehicles_BUB_OUT_Zona_Otras',
'vehicles_BUB_OUT_5_seats',
'vehicles_BUB_OUT_+5_seats',
'vehicles_BUB_OUT_-5_seats',
'vehicles_BUB_OUT_nan_seats',
'vehicles_BUB_OUT_101-200_CO2',
'vehicles_BUB_OUT_0-100_CO2',
'vehicles_BUB_OUT_nan_CO2',
'vehicles_BUB_OUT_201-300_CO2',
'vehicles_BUB_OUT_+300_CO2',
'vehicles_BUB_IN',
'vehicles_BUB_IN_Zona_Granada',
'vehicles_BUB_IN_Zona_Catalunia_y_Otras',
'vehicles_BUB_IN_Zona_Andalucia_no_GR',
'vehicles_BUB_IN_Extranjero',
'vehicles_BUB_IN_Zona_Comunidad_de_Madrid',
'vehicles_BUB_IN_Zona_Extremadura_y_Otras',
'vehicles_BUB_IN_Zona_Otras',
'vehicles_BUB_IN_5_seats',
'vehicles_BUB_IN_+5_seats',
'vehicles_BUB_IN_-5_seats',
'vehicles_BUB_IN_nan_seats',
'vehicles_BUB_IN_101-200_CO2',
'vehicles_BUB_IN_0-100_CO2',
'vehicles_BUB_IN_nan_CO2',
'vehicles_BUB_IN_201-300_CO2',
'vehicles_BUB_IN_+300_CO2']
```

[54]:
```
trafico_feb22_ago23.head()
- **Viabilidad:**
```

[54]:
```
                        date  vehicles_CAP_OUT  vehicles_CAP_OUT_Zona_Granada  \
0  2022-02-04 12:00:00+00:00                23                             11
1  2022-02-04 13:00:00+00:00                25                             11
2  2022-02-04 14:00:00+00:00                32                             20
3  2022-02-04 15:00:00+00:00                32                             20
4  2022-02-04 16:00:00+00:00                30                             11

   vehicles_CAP_OUT_Zona_Catalunia_y_Otras  \
0                                        1
1                                        1
2                                        1
3                                        1
4                                        1

   vehicles_CAP_OUT_Zona_Andalucia_no_GR  vehicles_CAP_OUT_Extranjero  \
```

```
0                                      2                            5
1                                      4                            7
2                                      6                            4
3                                      6                            2
4                                      8                            3

   vehicles_CAP_OUT_Zona_Comunidad_de_Madrid  \
0                                          2
1                                          2
2                                          1
3                                          2
4                                          3

   vehicles_CAP_OUT_Zona_Extremadura_y_Otras  vehicles_CAP_OUT_Zona_Otras  \
0                                          2                            0
1                                          0                            0
2                                          0                            0
3                                          1                            0
4                                          4                            0

   vehicles_CAP_OUT_5_seats  vehicles_CAP_OUT_+5_seats  \
0                        12                          1
1                        10                          5
2                        23                          3
3                        24                          4
4                        22                          2

   vehicles_CAP_OUT_-5_seats  vehicles_CAP_OUT_nan_seats  \
0                          5                           5
1                          3                           7
2                          2                           4
3                          2                           2
4                          3                           3

   vehicles_CAP_OUT_101-200_CO2  vehicles_CAP_OUT_0-100_CO2  \
0                             6                           1
1                             7                           1
2                            13                           2
3                             9                           4
4                            13                           4

   vehicles_CAP_OUT_nan_CO2  vehicles_CAP_OUT_201-300_CO2  \
0                        15                             1
1                        16                             1
2                        17                             0
3                        19                             0
4                        13                             0
```

```
   vehicles_CAP_OUT_+300_CO2  vehicles_CAP_IN  vehicles_CAP_IN_Zona_Granada  \
0                          0               22                            11
1                          0               43                            17
2                          0               39                            14
3                          0               26                            10
4                          0               21                            11

   vehicles_CAP_IN_Zona_Catalunia_y_Otras  \
0                                       3
1                                       1
2                                       2
3                                       1
4                                       1

   vehicles_CAP_IN_Zona_Andalucia_no_GR  vehicles_CAP_IN_Extranjero  \
0                                     6                           0
1                                    16                           4
2                                    10                           7
3                                     2                           7
4                                     5                           2

   vehicles_CAP_IN_Zona_Comunidad_de_Madrid  \
0                                         0
1                                         3
2                                         3
3                                         3
4                                         2

   vehicles_CAP_IN_Zona_Extremadura_y_Otras  vehicles_CAP_IN_Zona_Otras  \
0                                         2                           0
1                                         2                           0
2                                         3                           0
3                                         3                           0
4                                         0                           0

   vehicles_CAP_IN_5_seats  vehicles_CAP_IN_+5_seats  \
0                       17                         3
1                       36                         0
2                       21                         6
3                       15                         2
4                       14                         1

   vehicles_CAP_IN_-5_seats  vehicles_CAP_IN_nan_seats  \
0                         2                          0
1                         4                          3
2                         6                          6
```

```
3                        2                          7
4                        4                          2

   vehicles_CAP_IN_101-200_CO2  vehicles_CAP_IN_0-100_CO2  \
0                           12                          2
1                           16                          4
2                           12                          2
3                            7                          3
4                            5                          5

   vehicles_CAP_IN_nan_CO2  vehicles_CAP_IN_201-300_CO2  \
0                        8                            0
1                       23                            0
2                       25                            0
3                       16                            0
4                       11                            0

   vehicles_CAP_IN_+300_CO2  vehicles_PAM_1_OUT  \
0                         0                   0
1                         0                   0
2                         0                   0
3                         0                   0
4                         0                   0

   vehicles_PAM_1_OUT_Zona_Granada  vehicles_PAM_1_OUT_Zona_Catalunia_y_Otras  \
0                                0                                          0
1                                0                                          0
2                                0                                          0
3                                0                                          0
4                                0                                          0

   vehicles_PAM_1_OUT_Zona_Andalucia_no_GR  vehicles_PAM_1_OUT_Extranjero  \
0                                        0                              0
1                                        0                              0
2                                        0                              0
3                                        0                              0
4                                        0                              0

   vehicles_PAM_1_OUT_Zona_Comunidad_de_Madrid  \
0                                            0
1                                            0
2                                            0
3                                            0
4                                            0

   vehicles_PAM_1_OUT_Zona_Extremadura_y_Otras  vehicles_PAM_1_OUT_Zona_Otras  \
0                                            0                              0
```

```
1                                         0                             0
2                                         0                             0
3                                         0                             0
4                                         0                             0

   vehicles_PAM_1_OUT_5_seats  vehicles_PAM_1_OUT_+5_seats  \
0                           0                            0
1                           0                            0
2                           0                            0
3                           0                            0
4                           0                            0

   vehicles_PAM_1_OUT_-5_seats  vehicles_PAM_1_OUT_nan_seats  \
0                            0                             0
1                            0                             0
2                            0                             0
3                            0                             0
4                            0                             0

   vehicles_PAM_1_OUT_101-200_CO2  vehicles_PAM_1_OUT_0-100_CO2  \
0                               0                             0
1                               0                             0
2                               0                             0
3                               0                             0
4                               0                             0

   vehicles_PAM_1_OUT_nan_CO2  vehicles_PAM_1_OUT_201-300_CO2  \
0                           0                               0
1                           0                               0
2                           0                               0
3                           0                               0
4                           0                               0

   vehicles_PAM_1_OUT_+300_CO2  vehicles_PAM_1_IN  \
0                            0                 36
1                            0                 85
2                            0                 60
3                            0                 41
4                            0                 50

   vehicles_PAM_1_IN_Zona_Granada  vehicles_PAM_1_IN_Zona_Catalunia_y_Otras  \
0                              16                                         1
1                              38                                         4
2                              27                                         4
3                              21                                         1
4                              20                                         4
```

```
   vehicles_PAM_1_IN_Zona_Andalucia_no_GR  vehicles_PAM_1_IN_Extranjero  \
0                                       7                             8
1                                      22                             8
2                                       9                             7
3                                       0                             9
4                                       9                             9

   vehicles_PAM_1_IN_Zona_Comunidad_de_Madrid  \
0                                           2
1                                           6
2                                           7
3                                           5
4                                           4

   vehicles_PAM_1_IN_Zona_Extremadura_y_Otras  vehicles_PAM_1_IN_Zona_Otras  \
0                                           2                             0
1                                           7                             0
2                                           6                             0
3                                           5                             0
4                                           4                             0

   vehicles_PAM_1_IN_5_seats  vehicles_PAM_1_IN_+5_seats  \
0                         25                           1
1                         61                          11
2                         43                           4
3                         22                           4
4                         36                           3

   vehicles_PAM_1_IN_-5_seats  vehicles_PAM_1_IN_nan_seats  \
0                           3                            7
1                           6                            7
2                           7                            6
3                           6                            9
4                           3                            8

   vehicles_PAM_1_IN_101-200_CO2  vehicles_PAM_1_IN_0-100_CO2  \
0                             16                            2
1                             30                            6
2                             26                            4
3                             16                            1
4                             14                            4

   vehicles_PAM_1_IN_nan_CO2  vehicles_PAM_1_IN_201-300_CO2  \
0                         18                              0
1                         46                              3
2                         29                              1
3                         23                              1
```

```
4                         30                              2

   vehicles_PAM_1_IN_+300_CO2  vehicles_PAM_2_OUT  \
0                           0                   0
1                           0                   0
2                           0                   0
3                           0                   0
4                           0                   0

   vehicles_PAM_2_OUT_Zona_Granada  vehicles_PAM_2_OUT_Zona_Catalunia_y_Otras  \
0                                0                                          0
1                                0                                          0
2                                0                                          0
3                                0                                          0
4                                0                                          0

   vehicles_PAM_2_OUT_Zona_Andalucia_no_GR  vehicles_PAM_2_OUT_Extranjero  \
0                                        0                              0
1                                        0                              0
2                                        0                              0
3                                        0                              0
4                                        0                              0

   vehicles_PAM_2_OUT_Zona_Comunidad_de_Madrid  \
0                                            0
1                                            0
2                                            0
3                                            0
4                                            0

   vehicles_PAM_2_OUT_Zona_Extremadura_y_Otras  vehicles_PAM_2_OUT_Zona_Otras  \
0                                            0                              0
1                                            0                              0
2                                            0                              0
3                                            0                              0
4                                            0                              0

   vehicles_PAM_2_OUT_5_seats  vehicles_PAM_2_OUT_+5_seats  \
0                           0                            0
1                           0                            0
2                           0                            0
3                           0                            0
4                           0                            0

   vehicles_PAM_2_OUT_-5_seats  vehicles_PAM_2_OUT_nan_seats  \
0                            0                             0
1                            0                             0
```

```
2                          0                           0
3                          0                           0
4                          0                           0

   vehicles_PAM_2_OUT_101-200_CO2  vehicles_PAM_2_OUT_0-100_CO2  \
0                               0                             0
1                               0                             0
2                               0                             0
3                               0                             0
4                               0                             0

   vehicles_PAM_2_OUT_nan_CO2  vehicles_PAM_2_OUT_201-300_CO2  \
0                           0                               0
1                           0                               0
2                           0                               0
3                           0                               0
4                           0                               0

   vehicles_PAM_2_OUT_+300_CO2  vehicles_PAM_2_IN  \
0                            0                  0
1                            0                  0
2                            0                  0
3                            0                  0
4                            0                  0

   vehicles_PAM_2_IN_Zona_Granada  vehicles_PAM_2_IN_Zona_Catalunia_y_Otras  \
0                               0                                         0
1                               0                                         0
2                               0                                         0
3                               0                                         0
4                               0                                         0

   vehicles_PAM_2_IN_Zona_Andalucia_no_GR  vehicles_PAM_2_IN_Extranjero  \
0                                       0                             0
1                                       0                             0
2                                       0                             0
3                                       0                             0
4                                       0                             0

   vehicles_PAM_2_IN_Zona_Comunidad_de_Madrid  \
0                                           0
1                                           0
2                                           0
3                                           0
4                                           0

   vehicles_PAM_2_IN_Zona_Extremadura_y_Otras  vehicles_PAM_2_IN_Zona_Otras  \
```

```
0                                              0                               0
1                                              0                               0
2                                              0                               0
3                                              0                               0
4                                              0                               0

   vehicles_PAM_2_IN_5_seats  vehicles_PAM_2_IN_+5_seats  \
0                          0                           0
1                          0                           0
2                          0                           0
3                          0                           0
4                          0                           0

   vehicles_PAM_2_IN_-5_seats  vehicles_PAM_2_IN_nan_seats  \
0                           0                            0
1                           0                            0
2                           0                            0
3                           0                            0
4                           0                            0

   vehicles_PAM_2_IN_101-200_CO2  vehicles_PAM_2_IN_0-100_CO2  \
0                              0                            0
1                              0                            0
2                              0                            0
3                              0                            0
4                              0                            0

   vehicles_PAM_2_IN_nan_CO2  vehicles_PAM_2_IN_201-300_CO2  \
0                          0                              0
1                          0                              0
2                          0                              0
3                          0                              0
4                          0                              0

   vehicles_PAM_2_IN_+300_CO2  vehicles_BUB_OUT  \
0                           0                 0
1                           0                 0
2                           0                 0
3                           0                 0
4                           0                 0

   vehicles_BUB_OUT_Zona_Granada  vehicles_BUB_OUT_Zona_Catalunia_y_Otras  \
0                              0                                        0
1                              0                                        0
2                              0                                        0
3                              0                                        0
4                              0                                        0
```

```
   vehicles_BUB_OUT_Zona_Andalucia_no_GR  vehicles_BUB_OUT_Extranjero  \
0                                      0                            0
1                                      0                            0
2                                      0                            0
3                                      0                            0
4                                      0                            0

   vehicles_BUB_OUT_Zona_Comunidad_de_Madrid  \
0                                          0
1                                          0
2                                          0
3                                          0
4                                          0

   vehicles_BUB_OUT_Zona_Extremadura_y_Otras  vehicles_BUB_OUT_Zona_Otras  \
0                                          0                            0
1                                          0                            0
2                                          0                            0
3                                          0                            0
4                                          0                            0

   vehicles_BUB_OUT_5_seats  vehicles_BUB_OUT_+5_seats  \
0                         0                          0
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   vehicles_BUB_OUT_-5_seats  vehicles_BUB_OUT_nan_seats  \
0                          0                           0
1                          0                           0
2                          0                           0
3                          0                           0
4                          0                           0

   vehicles_BUB_OUT_101-200_CO2  vehicles_BUB_OUT_0-100_CO2  \
0                             0                           0
1                             0                           0
2                             0                           0
3                             0                           0
4                             0                           0

   vehicles_BUB_OUT_nan_CO2  vehicles_BUB_OUT_201-300_CO2  \
0                         0                             0
1                         0                             0
2                         0                             0
```

```
3                        0                            0
4                        0                            0

   vehicles_BUB_OUT_+300_CO2  vehicles_BUB_IN  vehicles_BUB_IN_Zona_Granada  \
0                          0                0                             0
1                          0                0                             0
2                          0                0                             0
3                          0                0                             0
4                          0                0                             0

   vehicles_BUB_IN_Zona_Catalunia_y_Otras  \
0                                       0
1                                       0
2                                       0
3                                       0
4                                       0

   vehicles_BUB_IN_Zona_Andalucia_no_GR  vehicles_BUB_IN_Extranjero  \
0                                     0                           0
1                                     0                           0
2                                     0                           0
3                                     0                           0
4                                     0                           0

   vehicles_BUB_IN_Zona_Comunidad_de_Madrid  \
0                                         0
1                                         0
2                                         0
3                                         0
4                                         0

   vehicles_BUB_IN_Zona_Extremadura_y_Otras  vehicles_BUB_IN_Zona_Otras  \
0                                         0                           0
1                                         0                           0
2                                         0                           0
3                                         0                           0
4                                         0                           0

   vehicles_BUB_IN_5_seats  vehicles_BUB_IN_+5_seats  \
0                        0                         0
1                        0                         0
2                        0                         0
3                        0                         0
4                        0                         0

   vehicles_BUB_IN_-5_seats  vehicles_BUB_IN_nan_seats  \
0                         0                          0
```

```
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   vehicles_BUB_IN_101-200_CO2  vehicles_BUB_IN_0-100_CO2  \
0                            0                          0
1                            0                          0
2                            0                          0
3                            0                          0
4                            0                          0

   vehicles_BUB_IN_nan_CO2  vehicles_BUB_IN_201-300_CO2  \
0                        0                            0
1                        0                            0
2                        0                            0
3                        0                            0
4                        0                            0

   vehicles_BUB_IN_+300_CO2
0                         0
1                         0
2                         0
3                         0
4                         0
```

[55]:
```
trafico_contamina_interseccion = pd.read_csv('trafico_contamina_intersección.
 ↪csv')

pprint(list(trafico_contamina_interseccion.columns))
```

```
['Date',
 'CO',
 'NO2',
 'NO',
 'O3',
 'SO2',
 'PM10',
 'eBC_tot',
 'eBC_ff',
 'eBC_bb',
 'TEMP',
 'RH',
 'WS',
 'WD',
 'PRES',
 'truck_pos',
 'eBC',
```

```
'vehicles_PAM_1_OUT',
'vehicles_PAM_1_OUT_Zona_Granada',
'vehicles_PAM_1_OUT_Zona_Comunidad_de_Madrid',
'vehicles_PAM_1_OUT_Extranjero',
'vehicles_PAM_1_OUT_Zona_Andalucia_no_GR',
'vehicles_PAM_1_OUT_Zona_Catalunia_y_Otras',
'vehicles_PAM_1_OUT_Zona_Extremadura_y_Otras',
'vehicles_PAM_1_OUT_Zona_Otras',
'vehicles_PAM_1_OUT_5_seats',
'vehicles_PAM_1_OUT_nan_seats',
'vehicles_PAM_1_OUT_+5_seats',
'vehicles_PAM_1_OUT_-5_seats',
'vehicles_PAM_1_OUT_101-200_CO2',
'vehicles_PAM_1_OUT_nan_CO2',
'vehicles_PAM_1_OUT_+300_CO2',
'vehicles_PAM_1_OUT_0-100_CO2',
'vehicles_PAM_1_OUT_201-300_CO2',
'vehicles_PAM_1_OUT_Etiqueta C',
'vehicles_PAM_1_OUT_Sin Etiqueta',
'vehicles_PAM_1_OUT_Etiqueta B',
'vehicles_PAM_1_OUT_Etiqueta ECO',
'vehicles_PAM_1_OUT_Etiqueta 0',
'vehicles_PAM_1_IN',
'vehicles_PAM_1_IN_Zona_Granada',
'vehicles_PAM_1_IN_Zona_Comunidad_de_Madrid',
'vehicles_PAM_1_IN_Extranjero',
'vehicles_PAM_1_IN_Zona_Andalucia_no_GR',
'vehicles_PAM_1_IN_Zona_Catalunia_y_Otras',
'vehicles_PAM_1_IN_Zona_Extremadura_y_Otras',
'vehicles_PAM_1_IN_Zona_Otras',
'vehicles_PAM_1_IN_5_seats',
'vehicles_PAM_1_IN_nan_seats',
'vehicles_PAM_1_IN_+5_seats',
'vehicles_PAM_1_IN_-5_seats',
'vehicles_PAM_1_IN_101-200_CO2',
'vehicles_PAM_1_IN_nan_CO2',
'vehicles_PAM_1_IN_+300_CO2',
'vehicles_PAM_1_IN_0-100_CO2',
'vehicles_PAM_1_IN_201-300_CO2',
'vehicles_PAM_1_IN_Etiqueta C',
'vehicles_PAM_1_IN_Sin Etiqueta',
'vehicles_PAM_1_IN_Etiqueta B',
'vehicles_PAM_1_IN_Etiqueta ECO',
'vehicles_PAM_1_IN_Etiqueta 0',
'vehicles_PAM_2_OUT',
'vehicles_PAM_2_OUT_Zona_Granada',
'vehicles_PAM_2_OUT_Zona_Comunidad_de_Madrid',
'vehicles_PAM_2_OUT_Extranjero',
```

```
'vehicles_PAM_2_OUT_Zona_Andalucia_no_GR',
'vehicles_PAM_2_OUT_Zona_Catalunia_y_Otras',
'vehicles_PAM_2_OUT_Zona_Extremadura_y_Otras',
'vehicles_PAM_2_OUT_Zona_Otras',
'vehicles_PAM_2_OUT_5_seats',
'vehicles_PAM_2_OUT_nan_seats',
'vehicles_PAM_2_OUT_+5_seats',
'vehicles_PAM_2_OUT_-5_seats',
'vehicles_PAM_2_OUT_101-200_CO2',
'vehicles_PAM_2_OUT_nan_CO2',
'vehicles_PAM_2_OUT_+300_CO2',
'vehicles_PAM_2_OUT_0-100_CO2',
'vehicles_PAM_2_OUT_201-300_CO2',
'vehicles_PAM_2_OUT_Etiqueta C',
'vehicles_PAM_2_OUT_Sin Etiqueta',
'vehicles_PAM_2_OUT_Etiqueta B',
'vehicles_PAM_2_OUT_Etiqueta ECO',
'vehicles_PAM_2_OUT_Etiqueta 0',
'vehicles_PAM_2_IN',
'vehicles_PAM_2_IN_Zona_Granada',
'vehicles_PAM_2_IN_Zona_Comunidad_de_Madrid',
'vehicles_PAM_2_IN_Extranjero',
'vehicles_PAM_2_IN_Zona_Andalucia_no_GR',
'vehicles_PAM_2_IN_Zona_Catalunia_y_Otras',
'vehicles_PAM_2_IN_Zona_Extremadura_y_Otras',
'vehicles_PAM_2_IN_Zona_Otras',
'vehicles_PAM_2_IN_5_seats',
'vehicles_PAM_2_IN_nan_seats',
'vehicles_PAM_2_IN_+5_seats',
'vehicles_PAM_2_IN_-5_seats',
'vehicles_PAM_2_IN_101-200_CO2',
'vehicles_PAM_2_IN_nan_CO2',
'vehicles_PAM_2_IN_+300_CO2',
'vehicles_PAM_2_IN_0-100_CO2',
'vehicles_PAM_2_IN_201-300_CO2',
'vehicles_PAM_2_IN_Etiqueta C',
'vehicles_PAM_2_IN_Sin Etiqueta',
'vehicles_PAM_2_IN_Etiqueta B',
'vehicles_PAM_2_IN_Etiqueta ECO',
'vehicles_PAM_2_IN_Etiqueta 0',
'vehicles_CAP_OUT',
'vehicles_CAP_OUT_Zona_Granada',
'vehicles_CAP_OUT_Zona_Comunidad_de_Madrid',
'vehicles_CAP_OUT_Extranjero',
'vehicles_CAP_OUT_Zona_Andalucia_no_GR',
'vehicles_CAP_OUT_Zona_Catalunia_y_Otras',
'vehicles_CAP_OUT_Zona_Extremadura_y_Otras',
'vehicles_CAP_OUT_Zona_Otras',
```

```
'vehicles_CAP_OUT_5_seats',
'vehicles_CAP_OUT_nan_seats',
'vehicles_CAP_OUT_+5_seats',
'vehicles_CAP_OUT_-5_seats',
'vehicles_CAP_OUT_101-200_CO2',
'vehicles_CAP_OUT_nan_CO2',
'vehicles_CAP_OUT_+300_CO2',
'vehicles_CAP_OUT_0-100_CO2',
'vehicles_CAP_OUT_201-300_CO2',
'vehicles_CAP_OUT_Etiqueta C',
'vehicles_CAP_OUT_Sin Etiqueta',
'vehicles_CAP_OUT_Etiqueta B',
'vehicles_CAP_OUT_Etiqueta ECO',
'vehicles_CAP_OUT_Etiqueta 0',
'vehicles_CAP_IN',
'vehicles_CAP_IN_Zona_Granada',
'vehicles_CAP_IN_Zona_Comunidad_de_Madrid',
'vehicles_CAP_IN_Extranjero',
'vehicles_CAP_IN_Zona_Andalucia_no_GR',
'vehicles_CAP_IN_Zona_Catalunia_y_Otras',
'vehicles_CAP_IN_Zona_Extremadura_y_Otras',
'vehicles_CAP_IN_Zona_Otras',
'vehicles_CAP_IN_5_seats',
'vehicles_CAP_IN_nan_seats',
'vehicles_CAP_IN_+5_seats',
'vehicles_CAP_IN_-5_seats',
'vehicles_CAP_IN_101-200_CO2',
'vehicles_CAP_IN_nan_CO2',
'vehicles_CAP_IN_+300_CO2',
'vehicles_CAP_IN_0-100_CO2',
'vehicles_CAP_IN_201-300_CO2',
'vehicles_CAP_IN_Etiqueta C',
'vehicles_CAP_IN_Sin Etiqueta',
'vehicles_CAP_IN_Etiqueta B',
'vehicles_CAP_IN_Etiqueta ECO',
'vehicles_CAP_IN_Etiqueta 0',
'vehicles_BUB_OUT',
'vehicles_BUB_OUT_Zona_Granada',
'vehicles_BUB_OUT_Zona_Comunidad_de_Madrid',
'vehicles_BUB_OUT_Extranjero',
'vehicles_BUB_OUT_Zona_Andalucia_no_GR',
'vehicles_BUB_OUT_Zona_Catalunia_y_Otras',
'vehicles_BUB_OUT_Zona_Extremadura_y_Otras',
'vehicles_BUB_OUT_Zona_Otras',
'vehicles_BUB_OUT_5_seats',
'vehicles_BUB_OUT_nan_seats',
'vehicles_BUB_OUT_+5_seats',
'vehicles_BUB_OUT_-5_seats',
```

```
 'vehicles_BUB_OUT_101-200_CO2',
 'vehicles_BUB_OUT_nan_CO2',
 'vehicles_BUB_OUT_+300_CO2',
 'vehicles_BUB_OUT_0-100_CO2',
 'vehicles_BUB_OUT_201-300_CO2',
 'vehicles_BUB_OUT_Etiqueta C',
 'vehicles_BUB_OUT_Sin Etiqueta',
 'vehicles_BUB_OUT_Etiqueta B',
 'vehicles_BUB_OUT_Etiqueta ECO',
 'vehicles_BUB_OUT_Etiqueta 0',
 'vehicles_BUB_IN',
 'vehicles_BUB_IN_Zona_Granada',
 'vehicles_BUB_IN_Zona_Comunidad_de_Madrid',
 'vehicles_BUB_IN_Extranjero',
 'vehicles_BUB_IN_Zona_Andalucia_no_GR',
 'vehicles_BUB_IN_Zona_Catalunia_y_Otras',
 'vehicles_BUB_IN_Zona_Extremadura_y_Otras',
 'vehicles_BUB_IN_Zona_Otras',
 'vehicles_BUB_IN_5_seats',
 'vehicles_BUB_IN_nan_seats',
 'vehicles_BUB_IN_+5_seats',
 'vehicles_BUB_IN_-5_seats',
 'vehicles_BUB_IN_101-200_CO2',
 'vehicles_BUB_IN_nan_CO2',
 'vehicles_BUB_IN_+300_CO2',
 'vehicles_BUB_IN_0-100_CO2',
 'vehicles_BUB_IN_201-300_CO2',
 'vehicles_BUB_IN_Etiqueta C',
 'vehicles_BUB_IN_Sin Etiqueta',
 'vehicles_BUB_IN_Etiqueta B',
 'vehicles_BUB_IN_Etiqueta ECO',
 'vehicles_BUB_IN_Etiqueta 0']
```

[56]:
```
trafico_contamina_interseccion.head()
```

[56]:

	Date	CO	NO2	NO	O3	SO2	PM10	eBC_tot	eBC_ff	\
0	2022-11-16 16:00:00	0.39	3.8	0.4	71.6	NaN	NaN	0.3	0.2	
1	2022-11-16 17:00:00	0.50	2.9	0.1	68.4	NaN	NaN	0.1	0.0	
2	2022-11-16 18:00:00	0.29	4.4	0.6	62.8	NaN	NaN	0.6	0.1	
3	2022-11-16 19:00:00	0.26	5.0	0.8	62.3	NaN	NaN	0.9	0.0	
4	2022-11-16 20:00:00	0.20	3.8	0.5	60.1	NaN	NaN	0.6	0.2	

	eBC_bb	TEMP	RH	WS	WD	PRES	truck_pos	eBC	vehicles_PAM_1_OUT	\
0	0.1	11.6	62.4	3.95	144.1	852.3	CAP	NaN	49	
1	0.1	9.8	84.3	3.42	158.6	852.7	CAP	NaN	20	
2	0.5	9.8	85.1	2.61	168.9	852.9	CAP	NaN	0	
3	0.9	10.7	78.8	2.49	178.4	853.3	CAP	NaN	0	

```
4     0.4   9.6  91.0  2.51  172.0  853.5       CAP  NaN                  0

   vehicles_PAM_1_OUT_Zona_Granada  \
0                               27
1                                8
2                                0
3                                0
4                                0

   vehicles_PAM_1_OUT_Zona_Comunidad_de_Madrid  vehicles_PAM_1_OUT_Extranjero  \
0                                            5                              3
1                                            2                              6
2                                            0                              0
3                                            0                              0
4                                            0                              0

   vehicles_PAM_1_OUT_Zona_Andalucia_no_GR  \
0                                        6
1                                        2
2                                        0
3                                        0
4                                        0

   vehicles_PAM_1_OUT_Zona_Catalunia_y_Otras  \
0                                          3
1                                          1
2                                          0
3                                          0
4                                          0

   vehicles_PAM_1_OUT_Zona_Extremadura_y_Otras  vehicles_PAM_1_OUT_Zona_Otras  \
0                                            5                              0
1                                            1                              0
2                                            0                              0
3                                            0                              0
4                                            0                              0

   vehicles_PAM_1_OUT_5_seats  vehicles_PAM_1_OUT_nan_seats  \
0                          30                             3
1                           9                             6
2                           0                             0
3                           0                             0
4                           0                             0

   vehicles_PAM_1_OUT_+5_seats  vehicles_PAM_1_OUT_-5_seats  \
0                            5                           11
1                            2                            3
```

```
2                            0                            0
3                            0                            0
4                            0                            0

   vehicles_PAM_1_OUT_101-200_CO2  vehicles_PAM_1_OUT_nan_CO2  \
0                              26                          20
1                               6                          10
2                               0                           0
3                               0                           0
4                               0                           0

   vehicles_PAM_1_OUT_+300_CO2  vehicles_PAM_1_OUT_0-100_CO2  \
0                            1                             2
1                            0                             3
2                            0                             0
3                            0                             0
4                            0                             0

   vehicles_PAM_1_OUT_201-300_CO2  vehicles_PAM_1_OUT_Etiqueta C  \
0                               0                             20
1                               1                              7
2                               0                              0
3                               0                              0
4                               0                              0

   vehicles_PAM_1_OUT_Sin Etiqueta  vehicles_PAM_1_OUT_Etiqueta B  \
0                               15                             12
1                                9                              2
2                                0                              0
3                                0                              0
4                                0                              0

   vehicles_PAM_1_OUT_Etiqueta ECO  vehicles_PAM_1_OUT_Etiqueta 0  \
0                                2                              0
1                                0                              2
2                                0                              0
3                                0                              0
4                                0                              0

   vehicles_PAM_1_IN  vehicles_PAM_1_IN_Zona_Granada  \
0                 26                              18
1                 23                              14
2                  0                               0
3                  0                               0
4                  0                               0

   vehicles_PAM_1_IN_Zona_Comunidad_de_Madrid  vehicles_PAM_1_IN_Extranjero  \
```

```
0                                          2                              0
1                                          2                              5
2                                          0                              0
3                                          0                              0
4                                          0                              0

   vehicles_PAM_1_IN_Zona_Andalucia_no_GR  \
0                                       3
1                                       2
2                                       0
3                                       0
4                                       0

   vehicles_PAM_1_IN_Zona_Catalunia_y_Otras  \
0                                         1
1                                         0
2                                         0
3                                         0
4                                         0

   vehicles_PAM_1_IN_Zona_Extremadura_y_Otras  vehicles_PAM_1_IN_Zona_Otras  \
0                                           2                             0
1                                           0                             0
2                                           0                             0
3                                           0                             0
4                                           0                             0

   vehicles_PAM_1_IN_5_seats  vehicles_PAM_1_IN_nan_seats  \
0                         22                            0
1                         16                            5
2                          0                            0
3                          0                            0
4                          0                            0

   vehicles_PAM_1_IN_+5_seats  vehicles_PAM_1_IN_-5_seats  \
0                           3                           1
1                           0                           2
2                           0                           0
3                           0                           0
4                           0                           0

   vehicles_PAM_1_IN_101-200_CO2  vehicles_PAM_1_IN_nan_CO2  \
0                             11                         11
1                              7                         14
2                              0                          0
3                              0                          0
4                              0                          0
```

```
   vehicles_PAM_1_IN_+300_CO2  vehicles_PAM_1_IN_0-100_CO2  \
0                           0                            2
1                           0                            2
2                           0                            0
3                           0                            0
4                           0                            0

   vehicles_PAM_1_IN_201-300_CO2  vehicles_PAM_1_IN_Etiqueta C  \
0                              2                             8
1                              0                             9
2                              0                             0
3                              0                             0
4                              0                             0

   vehicles_PAM_1_IN_Sin Etiqueta  vehicles_PAM_1_IN_Etiqueta B  \
0                               9                             7
1                              12                             2
2                               0                             0
3                               0                             0
4                               0                             0

   vehicles_PAM_1_IN_Etiqueta ECO  vehicles_PAM_1_IN_Etiqueta 0  \
0                               2                             0
1                               0                             0
2                               0                             0
3                               0                             0
4                               0                             0

   vehicles_PAM_2_OUT  vehicles_PAM_2_OUT_Zona_Granada  \
0                  31                               15
1                  27                               15
2                   3                                1
3                   0                                0
4                   0                                0

   vehicles_PAM_2_OUT_Zona_Comunidad_de_Madrid  vehicles_PAM_2_OUT_Extranjero  \
0                                            3                              2
1                                            1                              9
2                                            1                              1
3                                            0                              0
4                                            0                              0

   vehicles_PAM_2_OUT_Zona_Andalucia_no_GR  \
0                                        5
1                                        1
2                                        0
```

```
3                                      0
4                                      0

   vehicles_PAM_2_OUT_Zona_Catalunia_y_Otras  \
0                                          3
1                                          1
2                                          0
3                                          0
4                                          0

   vehicles_PAM_2_OUT_Zona_Extremadura_y_Otras  vehicles_PAM_2_OUT_Zona_Otras  \
0                                            3                              0
1                                            0                              0
2                                            0                              0
3                                            0                              0
4                                            0                              0

   vehicles_PAM_2_OUT_5_seats  vehicles_PAM_2_OUT_nan_seats  \
0                          22                             2
1                          12                             9
2                           2                             1
3                           0                             0
4                           0                             0

   vehicles_PAM_2_OUT_+5_seats  vehicles_PAM_2_OUT_-5_seats  \
0                            2                            5
1                            2                            4
2                            0                            0
3                            0                            0
4                            0                            0

   vehicles_PAM_2_OUT_101-200_CO2  vehicles_PAM_2_OUT_nan_CO2  \
0                              16                          12
1                               4                          19
2                               0                           3
3                               0                           0
4                               0                           0

   vehicles_PAM_2_OUT_+300_CO2  vehicles_PAM_2_OUT_0-100_CO2  \
0                            0                             3
1                            0                             3
2                            0                             0
3                            0                             0
4                            0                             0

   vehicles_PAM_2_OUT_201-300_CO2  vehicles_PAM_2_OUT_Etiqueta C  \
0                               0                             10
```

```
1                              1                              6
2                              0                              0
3                              0                              0
4                              0                              0

   vehicles_PAM_2_OUT_Sin Etiqueta  vehicles_PAM_2_OUT_Etiqueta B  \
0                                9                              7
1                               14                              7
2                                2                              1
3                                0                              0
4                                0                              0

   vehicles_PAM_2_OUT_Etiqueta ECO  vehicles_PAM_2_OUT_Etiqueta 0  \
0                                5                              0
1                                0                              0
2                                0                              0
3                                0                              0
4                                0                              0

   vehicles_PAM_2_IN  vehicles_PAM_2_IN_Zona_Granada  \
0                 46                              26
1                 28                              20
2                  1                               0
3                  0                               0
4                  0                               0

   vehicles_PAM_2_IN_Zona_Comunidad_de_Madrid  vehicles_PAM_2_IN_Extranjero  \
0                                           4                             5
1                                           4                             1
2                                           1                             0
3                                           0                             0
4                                           0                             0

   vehicles_PAM_2_IN_Zona_Andalucia_no_GR  \
0                                       7
1                                       2
2                                       0
3                                       0
4                                       0

   vehicles_PAM_2_IN_Zona_Catalunia_y_Otras  \
0                                         2
1                                         0
2                                         0
3                                         0
4                                         0
```

	vehicles_PAM_2_IN_Zona_Extremadura_y_Otras	vehicles_PAM_2_IN_Zona_Otras	\
0	2	0	
1	1	0	
2	0	0	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_5_seats	vehicles_PAM_2_IN_nan_seats	\
0	31	5	
1	22	1	
2	1	0	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_+5_seats	vehicles_PAM_2_IN_-5_seats	\
0	3	7	
1	4	1	
2	0	0	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_101-200_CO2	vehicles_PAM_2_IN_nan_CO2	\
0	21	22	
1	10	15	
2	0	1	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_+300_CO2	vehicles_PAM_2_IN_0-100_CO2	\
0	0	2	
1	0	3	
2	0	0	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_201-300_CO2	vehicles_PAM_2_IN_Etiqueta C	\
0	1	16	
1	0	8	
2	0	0	
3	0	0	
4	0	0	

	vehicles_PAM_2_IN_Sin Etiqueta	vehicles_PAM_2_IN_Etiqueta B	\
0	17	12	
1	7	9	
2	1	0	
3	0	0	

```
4                            0                            0

   vehicles_PAM_2_IN_Etiqueta ECO  vehicles_PAM_2_IN_Etiqueta 0  \
0                               1                             0
1                               2                             2
2                               0                             0
3                               0                             0
4                               0                             0

   vehicles_CAP_OUT  vehicles_CAP_OUT_Zona_Granada  \
0                23                             12
1                33                             25
2                14                              9
3                14                             11
4                 5                              4

   vehicles_CAP_OUT_Zona_Comunidad_de_Madrid  vehicles_CAP_OUT_Extranjero  \
0                                          1                            0
1                                          4                            2
2                                          0                            3
3                                          1                            1
4                                          0                            1

   vehicles_CAP_OUT_Zona_Andalucia_no_GR  \
0                                      6
1                                      0
2                                      1
3                                      0
4                                      0

   vehicles_CAP_OUT_Zona_Catalunia_y_Otras  \
0                                        2
1                                        0
2                                        1
3                                        0
4                                        0

   vehicles_CAP_OUT_Zona_Extremadura_y_Otras  vehicles_CAP_OUT_Zona_Otras  \
0                                          2                            0
1                                          2                            0
2                                          0                            0
3                                          1                            0
4                                          0                            0

   vehicles_CAP_OUT_5_seats  vehicles_CAP_OUT_nan_seats  \
0                        17                           0
1                        23                           2
```

```
2                    10                           3
3                    11                           1
4                     4                           1

   vehicles_CAP_OUT_+5_seats  vehicles_CAP_OUT_-5_seats  \
0                          3                          3
1                          2                          6
2                          0                          1
3                          1                          1
4                          0                          0

   vehicles_CAP_OUT_101-200_CO2  vehicles_CAP_OUT_nan_CO2  \
0                             9                        10
1                            12                        17
2                             2                        12
3                             5                         8
4                             3                         1

   vehicles_CAP_OUT_+300_CO2  vehicles_CAP_OUT_0-100_CO2  \
0                          1                           3
1                          0                           3
2                          0                           0
3                          0                           1
4                          0                           1

   vehicles_CAP_OUT_201-300_CO2  vehicles_CAP_OUT_Etiqueta C  \
0                             0                            8
1                             1                           12
2                             0                            3
3                             0                            5
4                             0                            0

   vehicles_CAP_OUT_Sin Etiqueta  vehicles_CAP_OUT_Etiqueta B  \
0                              9                            5
1                              8                           10
2                              8                            3
3                              5                            4
4                              1                            4

   vehicles_CAP_OUT_Etiqueta ECO  vehicles_CAP_OUT_Etiqueta 0  \
0                              0                            1
1                              2                            1
2                              0                            0
3                              0                            0
4                              0                            0

   vehicles_CAP_IN  vehicles_CAP_IN_Zona_Granada  \
```

```
0             22                             13  
1             24                             14  
2             17                             15  
3              6                              4  
4              2                              1  

   vehicles_CAP_IN_Zona_Comunidad_de_Madrid  vehicles_CAP_IN_Extranjero  \
0                                         2                           1   
1                                         3                           4   
2                                         1                           1   
3                                         0                           1   
4                                         0                           0   

   vehicles_CAP_IN_Zona_Andalucia_no_GR  \
0                                     2   
1                                     2   
2                                     0   
3                                     0   
4                                     1   

   vehicles_CAP_IN_Zona_Catalunia_y_Otras  \
0                                       2   
1                                       0   
2                                       0   
3                                       0   
4                                       0   

   vehicles_CAP_IN_Zona_Extremadura_y_Otras  vehicles_CAP_IN_Zona_Otras  \
0                                         2                           0   
1                                         1                           0   
2                                         0                           0   
3                                         1                           0   
4                                         0                           0   

   vehicles_CAP_IN_5_seats  vehicles_CAP_IN_nan_seats  \
0                       17                          1   
1                       16                          4   
2                       13                          1   
3                        3                          1   
4                        2                          0   

   vehicles_CAP_IN_+5_seats  vehicles_CAP_IN_-5_seats  \
0                         3                         1   
1                         2                         2   
2                         0                         3   
3                         0                         2   
4                         0                         0   
```

```
   vehicles_CAP_IN_101-200_CO2  vehicles_CAP_IN_nan_CO2  \
0                           12                        6
1                            7                       16
2                            6                        9
3                            2                        4
4                            0                        1

   vehicles_CAP_IN_+300_CO2  vehicles_CAP_IN_0-100_CO2  \
0                         0                          4
1                         1                          0
2                         0                          1
3                         0                          0
4                         0                          1

   vehicles_CAP_IN_201-300_CO2  vehicles_CAP_IN_Etiqueta C  \
0                            0                          10
1                            0                           7
2                            1                           6
3                            0                           2
4                            0                           0

   vehicles_CAP_IN_Sin Etiqueta  vehicles_CAP_IN_Etiqueta B  \
0                             5                           6
1                            11                           4
2                             6                           4
3                             3                           0
4                             1                           1

   vehicles_CAP_IN_Etiqueta ECO  vehicles_CAP_IN_Etiqueta 0  vehicles_BUB_OUT  \
0                             1                           0                23
1                             2                           0                 6
2                             1                           0                 0
3                             1                           0                 0
4                             0                           0                 0

   vehicles_BUB_OUT_Zona_Granada  vehicles_BUB_OUT_Zona_Comunidad_de_Madrid  \
0                              9                                          0
1                              3                                          0
2                              0                                          0
3                              0                                          0
4                              0                                          0

   vehicles_BUB_OUT_Extranjero  vehicles_BUB_OUT_Zona_Andalucia_no_GR  \
0                            6                                      5
1                            1                                      2
2                            0                                      0
```

```
3                           0                                      0
4                           0                                      0

   vehicles_BUB_OUT_Zona_Catalunia_y_Otras  \
0                                        1
1                                        0
2                                        0
3                                        0
4                                        0

   vehicles_BUB_OUT_Zona_Extremadura_y_Otras  vehicles_BUB_OUT_Zona_Otras  \
0                                          2                            0
1                                          0                            0
2                                          0                            0
3                                          0                            0
4                                          0                            0

   vehicles_BUB_OUT_5_seats  vehicles_BUB_OUT_nan_seats  \
0                        12                           6
1                         5                           1
2                         0                           0
3                         0                           0
4                         0                           0

   vehicles_BUB_OUT_+5_seats  vehicles_BUB_OUT_-5_seats  \
0                          1                          4
1                          0                          0
2                          0                          0
3                          0                          0
4                          0                          0

   vehicles_BUB_OUT_101-200_CO2  vehicles_BUB_OUT_nan_CO2  \
0                             9                        11
1                             3                         2
2                             0                         0
3                             0                         0
4                             0                         0

   vehicles_BUB_OUT_+300_CO2  vehicles_BUB_OUT_0-100_CO2  \
0                          1                           2
1                          0                           1
2                          0                           0
3                          0                           0
4                          0                           0

   vehicles_BUB_OUT_201-300_CO2  vehicles_BUB_OUT_Etiqueta C  \
0                             0                            8
```

```
1                             0                            2
2                             0                            0
3                             0                            0
4                             0                            0

   vehicles_BUB_OUT_Sin Etiqueta  vehicles_BUB_OUT_Etiqueta B  \
0                              9                            4
1                              2                            1
2                              0                            0
3                              0                            0
4                              0                            0

   vehicles_BUB_OUT_Etiqueta ECO  vehicles_BUB_OUT_Etiqueta 0  \
0                              1                            1
1                              0                            1
2                              0                            0
3                              0                            0
4                              0                            0

   vehicles_BUB_IN  vehicles_BUB_IN_Zona_Granada  \
0               26                            12
1               16                             8
2                0                             0
3                0                             0
4                0                             0

   vehicles_BUB_IN_Zona_Comunidad_de_Madrid  vehicles_BUB_IN_Extranjero  \
0                                         3                           4
1                                         3                           3
2                                         0                           0
3                                         0                           0
4                                         0                           0

   vehicles_BUB_IN_Zona_Andalucia_no_GR  \
0                                     4
1                                     1
2                                     0
3                                     0
4                                     0

   vehicles_BUB_IN_Zona_Catalunia_y_Otras  \
0                                       1
1                                       0
2                                       0
3                                       0
4                                       0
```

```
   vehicles_BUB_IN_Zona_Extremadura_y_Otras  vehicles_BUB_IN_Zona_Otras  \
0                                         2                           0
1                                         1                           0
2                                         0                           0
3                                         0                           0
4                                         0                           0

   vehicles_BUB_IN_5_seats  vehicles_BUB_IN_nan_seats  \
0                       19                          4
1                       12                          2
2                        0                          0
3                        0                          0
4                        0                          0

   vehicles_BUB_IN_+5_seats  vehicles_BUB_IN_-5_seats  \
0                         1                         2
1                         1                         1
2                         0                         0
3                         0                         0
4                         0                         0

   vehicles_BUB_IN_101-200_CO2  vehicles_BUB_IN_nan_CO2  \
0                           10                       11
1                            6                        8
2                            0                        0
3                            0                        0
4                            0                        0

   vehicles_BUB_IN_+300_CO2  vehicles_BUB_IN_0-100_CO2  \
0                         1                          4
1                         0                          1
2                         0                          0
3                         0                          0
4                         0                          0

   vehicles_BUB_IN_201-300_CO2  vehicles_BUB_IN_Etiqueta C  \
0                            0                           8
1                            1                           5
2                            0                           0
3                            0                           0
4                            0                           0

   vehicles_BUB_IN_Sin Etiqueta  vehicles_BUB_IN_Etiqueta B  \
0                             6                           8
1                             7                           3
2                             0                           0
3                             0                           0
```

```
4                              0                            0

   vehicles_BUB_IN_Etiqueta ECO  vehicles_BUB_IN_Etiqueta 0
0                              3                            1
1                              1                            0
2                              0                            0
3                              0                            0
4                              0                            0
```

[57]:
```
set(trafico_contamina_interseccion["truck_pos"])
```

[57]:
```
{'CAP', 'PAM_2'}
```

1.1 Periodo del camión de meteorología en las regiones

1.1.1 Capileira

- 16/11/2022-15/01/2023

1.1.2 Pampaneira

- 17/01/2023-14/03/2023
- 06/06/2023-27/06/2023

[58]:
```
trafico_contamina_interseccion["Date"] = pd.
 ↪to_datetime(trafico_contamina_interseccion["Date"])
```

[59]:
```
fechas_cap =␣
 ↪trafico_contamina_interseccion[trafico_contamina_interseccion["truck_pos"]␣
 ↪== "CAP"]["Date"]
trafico_cap =␣
 ↪trafico_contamina_interseccion[trafico_contamina_interseccion["truck_pos"]␣
 ↪== "CAP"]
trafico_pam =␣
 ↪trafico_contamina_interseccion[trafico_contamina_interseccion["truck_pos"]␣
 ↪== "PAM_2"]

fecha_filtro = pd.to_datetime("2023-06-06")
trafico_pam_periodo1 = trafico_pam[trafico_pam["Date"] < fecha_filtro]
trafico_pam_periodo2 = trafico_pam[trafico_pam["Date"] >= fecha_filtro]

fechas_pam_periodo1 = trafico_pam_periodo1["Date"]
fechas_pam_periodo2 = trafico_pam_periodo2["Date"]
```

[60]:
```
from datetime import datetime, timedelta

def obtener_dias_faltantes(fecha_lista):
```

```
    # Convertir las cadenas de fecha a objetos datetime y obtener solo las␣
↪fechas
    fechas = [fecha.date() for fecha in fecha_lista]

    # Determinar el rango de fechas
    fecha_inicio = min(fechas)
    fecha_fin = max(fechas)

    # Generar todos los días en el rango
    todos_los_dias = set(fecha_inicio + timedelta(days=i) for i in␣
↪range((fecha_fin - fecha_inicio).days + 1))

    # Convertir la lista de fechas a un conjunto para la comparación
    fechas_set = set(fechas)

    # Calcular los días faltantes
    dias_faltantes = todos_los_dias - fechas_set

    return sorted(dias_faltantes)
```

[61]:
```
dias_faltantes_cap = obtener_dias_faltantes(fechas_cap)
dias_faltantes_pam_periodo1 = obtener_dias_faltantes(fechas_pam_periodo1)
dias_faltantes_pam_periodo2 = obtener_dias_faltantes(fechas_pam_periodo2)

print("Capileira: 16/11/2022-15/01/2023")
print("Días faltantes: ")
for dia_faltante_cap in dias_faltantes_cap:
    print(f"* {dia_faltante_cap}")

print("\nPampaneira: 17/01/2023-14/03/2023")
print("Días faltantes: ")
for dia_faltante_pam in dias_faltantes_pam_periodo1:
    print(f"* {dia_faltante_pam}")

print("\nPampaneira: 06/06/2023-27/06/2023")
print("Días faltantes: ")
for dia_faltante_pam in dias_faltantes_pam_periodo2:
    print(f"* {dia_faltante_pam}")
```

```
Capileira: 16/11/2022-15/01/2023
Días faltantes:
* 2022-12-30
* 2022-12-31
* 2023-01-01
* 2023-01-02
* 2023-01-03
* 2023-01-04
* 2023-01-05
```

```
* 2023-01-06
* 2023-01-07
* 2023-01-08
* 2023-01-09
* 2023-01-10

Pampaneira: 17/01/2023-14/03/2023
Días faltantes:

Pampaneira: 06/06/2023-27/06/2023
Días faltantes:
* 2023-06-11
```

[63]:

```python
import pandas as pd

mean_values = trafico_contamina_interseccion.select_dtypes(include='number').
 ↪mean()

# Ajustar la configuración para mostrar todas las filas y columnas
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
print("Medias - Columnas: ")
print(mean_values)
```

```
Medias - Columnas:
CO                                            0.192576
NO2                                           4.153839
NO                                            0.765554
O3                                           74.185321
SO2                                           0.855969
PM10                                         18.369377
eBC_tot                                       0.489836
eBC_ff                                        0.325452
eBC_bb                                        0.132579
TEMP                                         10.614455
RH                                           58.498436
WS                                            1.611853
WD                                          157.230044
PRES                                        877.623671
eBC                                           0.442909
vehicles_PAM_1_OUT                            9.714745
vehicles_PAM_1_OUT_Zona_Granada               4.366559
vehicles_PAM_1_OUT_Zona_Comunidad_de_Madrid   0.829582
vehicles_PAM_1_OUT_Extranjero                 1.499770
vehicles_PAM_1_OUT_Zona_Andalucia_no_GR       1.686266
vehicles_PAM_1_OUT_Zona_Catalunia_y_Otras     0.671566
vehicles_PAM_1_OUT_Zona_Extremadura_y_Otras   0.648599
vehicles_PAM_1_OUT_Zona_Otras                 0.012402
```

vehicles_PAM_1_OUT_5_seats	6.543868
vehicles_PAM_1_OUT_nan_seats	1.391364
vehicles_PAM_1_OUT_+5_seats	0.644006
vehicles_PAM_1_OUT_-5_seats	1.135508
vehicles_PAM_1_OUT_101-200_CO2	4.053744
vehicles_PAM_1_OUT_nan_CO2	4.694074
vehicles_PAM_1_OUT_+300_CO2	0.003675
vehicles_PAM_1_OUT_0-100_CO2	0.768948
vehicles_PAM_1_OUT_201-300_CO2	0.194304
vehicles_PAM_1_OUT_Etiqueta C	3.491043
vehicles_PAM_1_OUT_Sin Etiqueta	3.214056
vehicles_PAM_1_OUT_Etiqueta B	2.569591
vehicles_PAM_1_OUT_Etiqueta ECO	0.335324
vehicles_PAM_1_OUT_Etiqueta 0	0.104731
vehicles_PAM_1_IN	29.823610
vehicles_PAM_1_IN_Zona_Granada	15.046853
vehicles_PAM_1_IN_Zona_Comunidad_de_Madrid	2.795590
vehicles_PAM_1_IN_Extranjero	2.677997
vehicles_PAM_1_IN_Zona_Andalucia_no_GR	6.245292
vehicles_PAM_1_IN_Zona_Catalunia_y_Otras	1.354616
vehicles_PAM_1_IN_Zona_Extremadura_y_Otras	1.656867
vehicles_PAM_1_IN_Zona_Otras	0.046394
vehicles_PAM_1_IN_5_seats	22.263206
vehicles_PAM_1_IN_nan_seats	2.460726
vehicles_PAM_1_IN_+5_seats	2.466238
vehicles_PAM_1_IN_-5_seats	2.633441
vehicles_PAM_1_IN_101-200_CO2	14.020211
vehicles_PAM_1_IN_nan_CO2	12.731741
vehicles_PAM_1_IN_+300_CO2	0.016537
vehicles_PAM_1_IN_0-100_CO2	2.436380
vehicles_PAM_1_IN_201-300_CO2	0.618741
vehicles_PAM_1_IN_Etiqueta C	11.815802
vehicles_PAM_1_IN_Sin Etiqueta	7.731741
vehicles_PAM_1_IN_Etiqueta B	8.632981
vehicles_PAM_1_IN_Etiqueta ECO	1.339458
vehicles_PAM_1_IN_Etiqueta 0	0.303629
vehicles_PAM_2_OUT	20.465319
vehicles_PAM_2_OUT_Zona_Granada	11.156178
vehicles_PAM_2_OUT_Zona_Comunidad_de_Madrid	1.791915
vehicles_PAM_2_OUT_Extranjero	1.985760
vehicles_PAM_2_OUT_Zona_Andalucia_no_GR	3.476803
vehicles_PAM_2_OUT_Zona_Catalunia_y_Otras	0.996785
vehicles_PAM_2_OUT_Zona_Extremadura_y_Otras	1.029858
vehicles_PAM_2_OUT_Zona_Otras	0.028020
vehicles_PAM_2_OUT_5_seats	14.321543
vehicles_PAM_2_OUT_nan_seats	1.871842
vehicles_PAM_2_OUT_+5_seats	1.663757
vehicles_PAM_2_OUT_-5_seats	2.608176

vehicles_PAM_2_OUT_101-200_CO2	8.955443
vehicles_PAM_2_OUT_nan_CO2	9.394580
vehicles_PAM_2_OUT_+300_CO2	0.009646
vehicles_PAM_2_OUT_0-100_CO2	1.657786
vehicles_PAM_2_OUT_201-300_CO2	0.447864
vehicles_PAM_2_OUT_Etiqueta C	7.660083
vehicles_PAM_2_OUT_Sin Etiqueta	6.067983
vehicles_PAM_2_OUT_Etiqueta B	5.826367
vehicles_PAM_2_OUT_Etiqueta ECO	0.774920
vehicles_PAM_2_OUT_Etiqueta 0	0.135967
vehicles_PAM_2_IN	22.631144
vehicles_PAM_2_IN_Zona_Granada	12.589343
vehicles_PAM_2_IN_Zona_Comunidad_de_Madrid	1.859440
vehicles_PAM_2_IN_Extranjero	1.968764
vehicles_PAM_2_IN_Zona_Andalucia_no_GR	4.304088
vehicles_PAM_2_IN_Zona_Catalunia_y_Otras	0.831419
vehicles_PAM_2_IN_Zona_Extremadura_y_Otras	1.050528
vehicles_PAM_2_IN_Zona_Otras	0.027561
vehicles_PAM_2_IN_5_seats	17.110703
vehicles_PAM_2_IN_nan_seats	1.868167
vehicles_PAM_2_IN_+5_seats	1.764814
vehicles_PAM_2_IN_-5_seats	1.887460
vehicles_PAM_2_IN_101-200_CO2	10.185117
vehicles_PAM_2_IN_nan_CO2	10.261828
vehicles_PAM_2_IN_+300_CO2	0.012402
vehicles_PAM_2_IN_0-100_CO2	1.719339
vehicles_PAM_2_IN_201-300_CO2	0.452458
vehicles_PAM_2_IN_Etiqueta C	8.585668
vehicles_PAM_2_IN_Sin Etiqueta	6.361047
vehicles_PAM_2_IN_Etiqueta B	6.607258
vehicles_PAM_2_IN_Etiqueta ECO	0.894350
vehicles_PAM_2_IN_Etiqueta 0	0.182820
vehicles_CAP_OUT	19.808452
vehicles_CAP_OUT_Zona_Granada	10.985301
vehicles_CAP_OUT_Zona_Comunidad_de_Madrid	1.749196
vehicles_CAP_OUT_Extranjero	1.322003
vehicles_CAP_OUT_Zona_Andalucia_no_GR	4.104731
vehicles_CAP_OUT_Zona_Catalunia_y_Otras	0.692696
vehicles_CAP_OUT_Zona_Extremadura_y_Otras	0.938907
vehicles_CAP_OUT_Zona_Otras	0.015618
vehicles_CAP_OUT_5_seats	14.568213
vehicles_CAP_OUT_nan_seats	1.220487
vehicles_CAP_OUT_+5_seats	1.492421
vehicles_CAP_OUT_-5_seats	2.527331
vehicles_CAP_OUT_101-200_CO2	8.451998
vehicles_CAP_OUT_nan_CO2	9.310060
vehicles_CAP_OUT_+300_CO2	0.021130
vehicles_CAP_OUT_0-100_CO2	1.527791

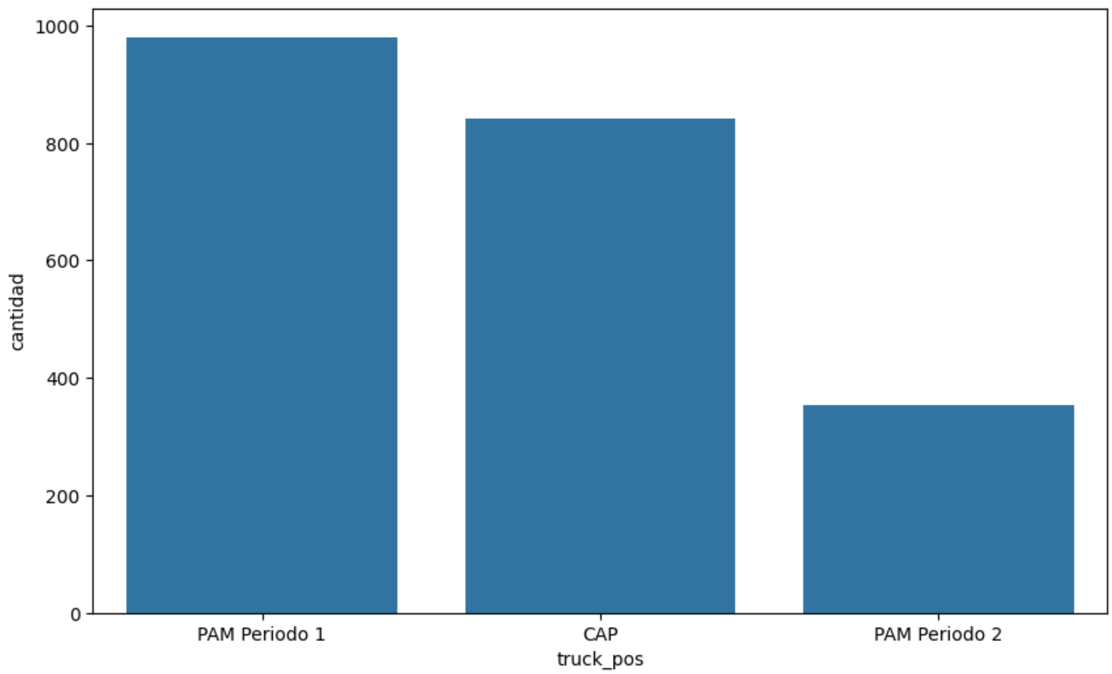
vehicles_CAP_OUT_201-300_CO2	0.497474
vehicles_CAP_OUT_Etiqueta C	7.308222
vehicles_CAP_OUT_Sin Etiqueta	6.020671
vehicles_CAP_OUT_Etiqueta B	5.612311
vehicles_CAP_OUT_Etiqueta ECO	0.761139
vehicles_CAP_OUT_Etiqueta 0	0.106109
vehicles_CAP_IN	21.503445
vehicles_CAP_IN_Zona_Granada	11.984842
vehicles_CAP_IN_Zona_Comunidad_de_Madrid	1.960037
vehicles_CAP_IN_Extranjero	1.013780
vehicles_CAP_IN_Zona_Andalucia_no_GR	4.601286
vehicles_CAP_IN_Zona_Catalunia_y_Otras	0.880570
vehicles_CAP_IN_Zona_Extremadura_y_Otras	1.045475
vehicles_CAP_IN_Zona_Otras	0.017455
vehicles_CAP_IN_5_seats	15.806155
vehicles_CAP_IN_nan_seats	0.902618
vehicles_CAP_IN_+5_seats	1.887919
vehicles_CAP_IN_-5_seats	2.906752
vehicles_CAP_IN_101-200_CO2	9.308682
vehicles_CAP_IN_nan_CO2	9.751952
vehicles_CAP_IN_+300_CO2	0.017455
vehicles_CAP_IN_0-100_CO2	1.881029
vehicles_CAP_IN_201-300_CO2	0.544327
vehicles_CAP_IN_Etiqueta C	8.248966
vehicles_CAP_IN_Sin Etiqueta	6.209003
vehicles_CAP_IN_Etiqueta B	6.106109
vehicles_CAP_IN_Etiqueta ECO	0.830041
vehicles_CAP_IN_Etiqueta 0	0.109325
vehicles_BUB_OUT	13.294901
vehicles_BUB_OUT_Zona_Granada	4.763436
vehicles_BUB_OUT_Zona_Comunidad_de_Madrid	1.150207
vehicles_BUB_OUT_Extranjero	2.853468
vehicles_BUB_OUT_Zona_Andalucia_no_GR	2.465319
vehicles_BUB_OUT_Zona_Catalunia_y_Otras	1.051906
vehicles_BUB_OUT_Zona_Extremadura_y_Otras	0.992650
vehicles_BUB_OUT_Zona_Otras	0.017915
vehicles_BUB_OUT_5_seats	8.107487
vehicles_BUB_OUT_nan_seats	2.692237
vehicles_BUB_OUT_+5_seats	1.012862
vehicles_BUB_OUT_-5_seats	1.482315
vehicles_BUB_OUT_101-200_CO2	5.581534
vehicles_BUB_OUT_nan_CO2	6.321543
vehicles_BUB_OUT_+300_CO2	0.011024
vehicles_BUB_OUT_0-100_CO2	1.098300
vehicles_BUB_OUT_201-300_CO2	0.282499
vehicles_BUB_OUT_Etiqueta C	4.996785
vehicles_BUB_OUT_Sin Etiqueta	4.686725
vehicles_BUB_OUT_Etiqueta B	3.052825

```
vehicles_BUB_OUT_Etiqueta ECO                       0.477722
vehicles_BUB_OUT_Etiqueta 0                         0.080845
vehicles_BUB_IN                                    18.207166
vehicles_BUB_IN_Zona_Granada                        8.375287
vehicles_BUB_IN_Zona_Comunidad_de_Madrid            1.876435
vehicles_BUB_IN_Extranjero                          1.840606
vehicles_BUB_IN_Zona_Andalucia_no_GR                4.096922
vehicles_BUB_IN_Zona_Catalunia_y_Otras              0.894350
vehicles_BUB_IN_Zona_Extremadura_y_Otras            1.097382
vehicles_BUB_IN_Zona_Otras                          0.026183
vehicles_BUB_IN_5_seats                            13.403307
vehicles_BUB_IN_nan_seats                           1.718420
vehicles_BUB_IN_+5_seats                            1.462104
vehicles_BUB_IN_-5_seats                            1.623335
vehicles_BUB_IN_101-200_CO2                         8.245751
vehicles_BUB_IN_nan_CO2                             8.093248
vehicles_BUB_IN_+300_CO2                            0.013780
vehicles_BUB_IN_0-100_CO2                           1.474047
vehicles_BUB_IN_201-300_CO2                         0.380340
vehicles_BUB_IN_Etiqueta C                          7.118052
vehicles_BUB_IN_Sin Etiqueta                        5.191548
vehicles_BUB_IN_Etiqueta B                          4.936610
vehicles_BUB_IN_Etiqueta ECO                        0.833257
vehicles_BUB_IN_Etiqueta 0                          0.127699
dtype: float64
```

1.1.3 Realizar interpolación para datos concretos de horas

1.2 Cantidad de elementos por truck_pos

[64]:
```
# Crear un DataFrame combinado
trafico_combined = pd.DataFrame({
    'truck_pos': ['PAM Periodo 1'] * len(trafico_pam_periodo1) + ['PAM Periodo␣
 ↪2'] * len(trafico_pam_periodo2) + ['CAP'] * len(trafico_cap),
    'Date': pd.concat([trafico_pam_periodo1['Date'],␣
 ↪trafico_pam_periodo2['Date'], trafico_cap['Date']])
})

# Contar los elementos por tipo de camión
counts = trafico_combined['truck_pos'].value_counts().reset_index()
counts.columns = ['truck_pos', 'cantidad']

# Crear el gráfico de barras
plt.figure(figsize=(10, 6))
sns.barplot(x='truck_pos', y='cantidad', data=counts)
plt.show()
```



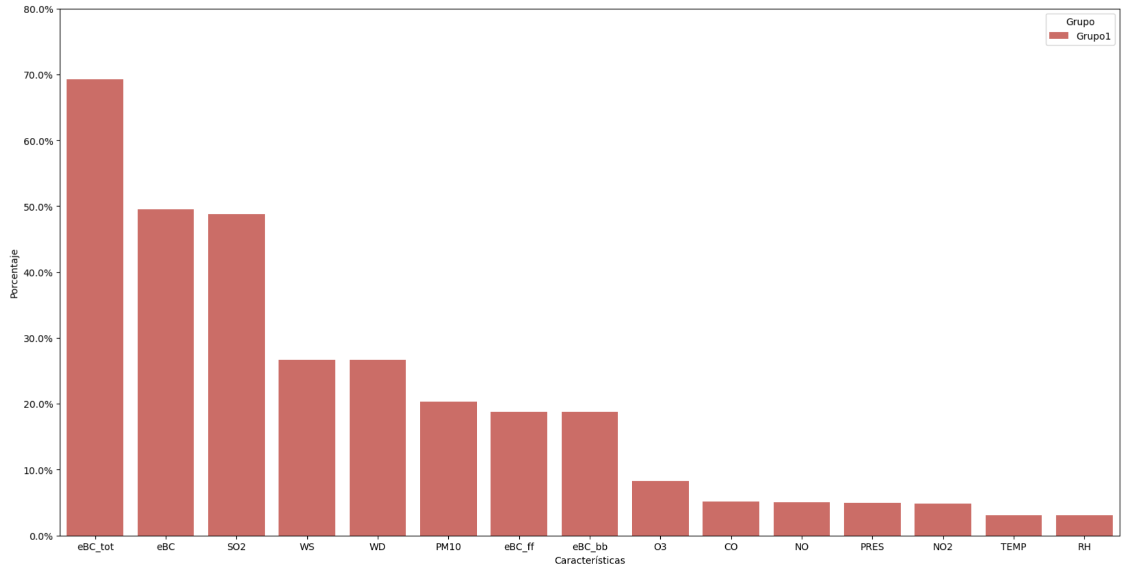
TODO: Interpolación para completar los valores nulos.

Se puede aprovechar teoría de Métodos Numéricos -> Splines

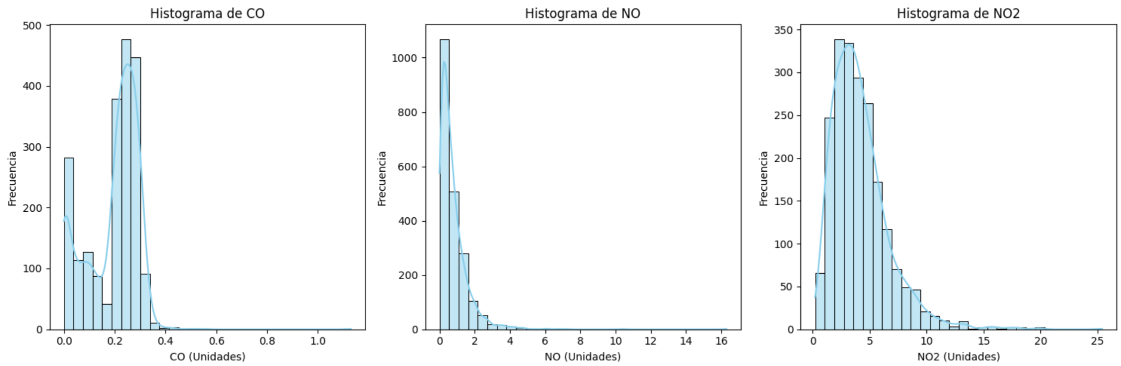
1.3 Porcentajes de valores nulos en cada columna

[65]:
```python
# Se crea un diccionario con clave el nombre de las columnas,
# y values el porcentaje de datos perdidos
nan_count = {k:list(trafico_contamina_interseccion.isna().sum()*100/
 ↪trafico_contamina_interseccion.shape[0])[i] for i,k in␣
 ↪enumerate(trafico_contamina_interseccion.columns)}

# Se ordena el diccionario en orden descendente
nan_count = {k: v for k, v in sorted(nan_count.items(), key=lambda item:␣
 ↪item[1],reverse=True)}

# Gráfico de barras
plt.figure(figsize=(20, 10))

# Crear un DataFrame con un grupo ficticio para el hue
data = pd.DataFrame({
    'Características': list(nan_count.keys())[:15],
    'Porcentaje': list(nan_count.values())[:15],
    'Grupo': ['Grupo1'] * 15  # Ficticio
})
```

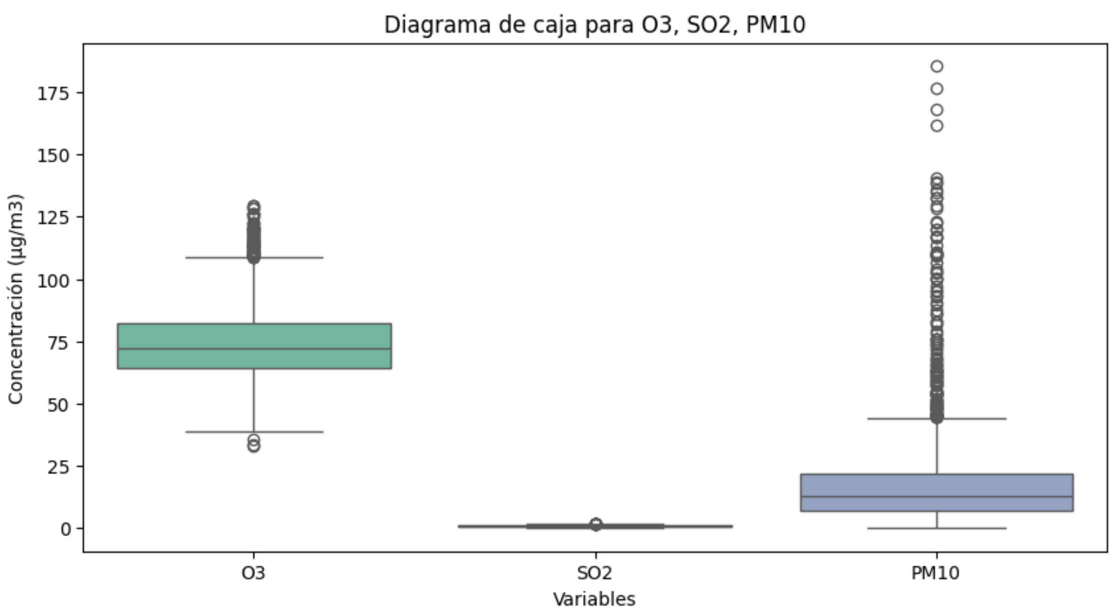
```
# Crear gráfico de barras con hue
plot = sns.barplot(x='Características', y='Porcentaje', hue='Grupo', data=data,␣
↪palette="hls")

# Establecer ticks y etiquetas de porcentaje
yticks = plot.get_yticks()  # Obtener ticks actuales
plot.set_yticks(yticks)  # Establecer ticks
plot.set_yticklabels(['{:.1f}%'.format(y) for y in yticks])  # Formatear␣
↪etiquetas

plt.show()
```



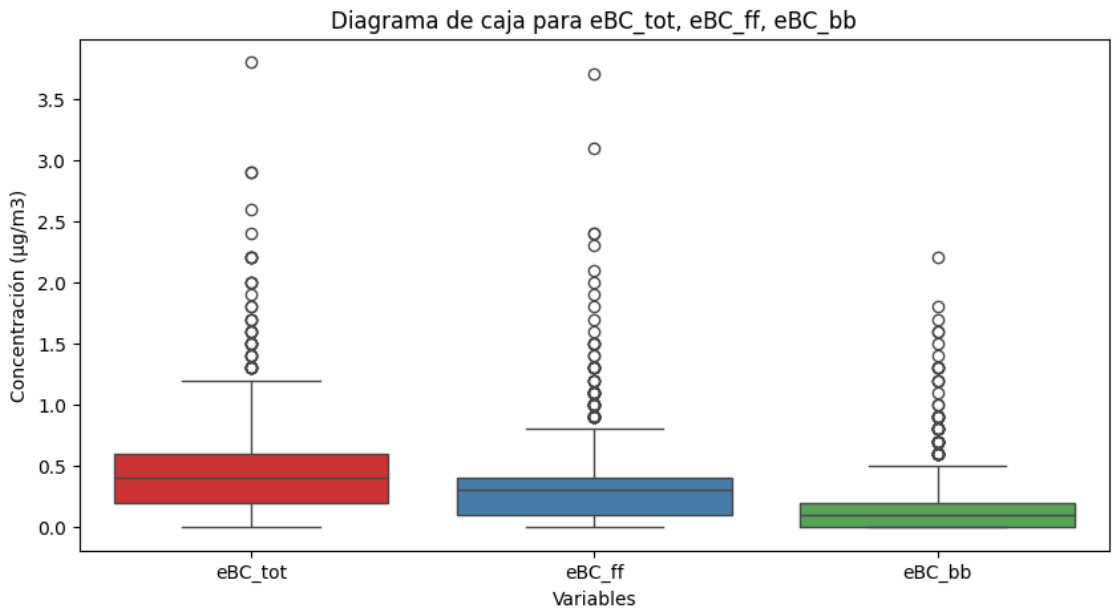
[66]:
```
variables_contaminacion = ['CO', 'NO', 'NO2']

# Histograma para CO, NO, NO2
plt.figure(figsize=(15, 5))
for i, variable in enumerate(variables_contaminacion):
    plt.subplot(1, 3, i+1)
    sns.histplot(trafico_contamina_interseccion[variable], bins=30, kde=True,␣
↪color='skyblue')
    plt.title(f'Histograma de {variable}')
    plt.xlabel(f'{variable} (Unidades)')
    plt.ylabel('Frecuencia')

plt.tight_layout()
plt.show()
```



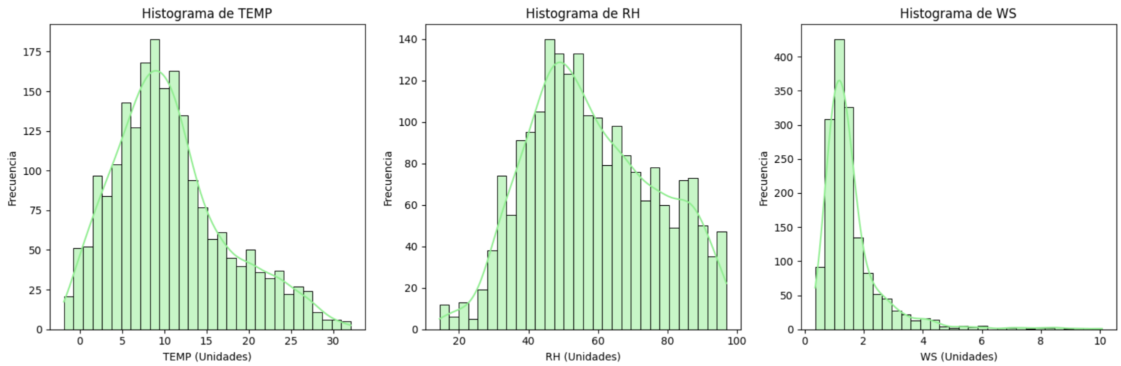
[67]:
```
# Variables de contaminación (ejemplo de DataFrame `df`)
# O3, SO2, PM10 deben ser columnas en el DataFrame
variables_contaminacion_boxplot = ['O3', 'SO2', 'PM10']

# Diagrama de caja para O3, SO2, PM10
plt.figure(figsize=(10, 5))
sns.
 ↪boxplot(data=trafico_contamina_interseccion[variables_contaminacion_boxplot],␣
 ↪palette='Set2')
plt.title('Diagrama de caja para O3, SO2, PM10')
plt.xlabel('Variables')
plt.ylabel('Concentración (µg/m3)')
plt.show()
```



[68]:
```
variables_contaminacion_ebc = ['eBC_tot', 'eBC_ff', 'eBC_bb']

# Diagrama de caja para eBC variables
plt.figure(figsize=(10, 5))
sns.boxplot(data=trafico_contamina_interseccion[variables_contaminacion_ebc],␣
↪palette='Set1')
plt.title('Diagrama de caja para eBC_tot, eBC_ff, eBC_bb')
plt.xlabel('Variables')
plt.ylabel('Concentración (µg/m3)')
plt.show()
```

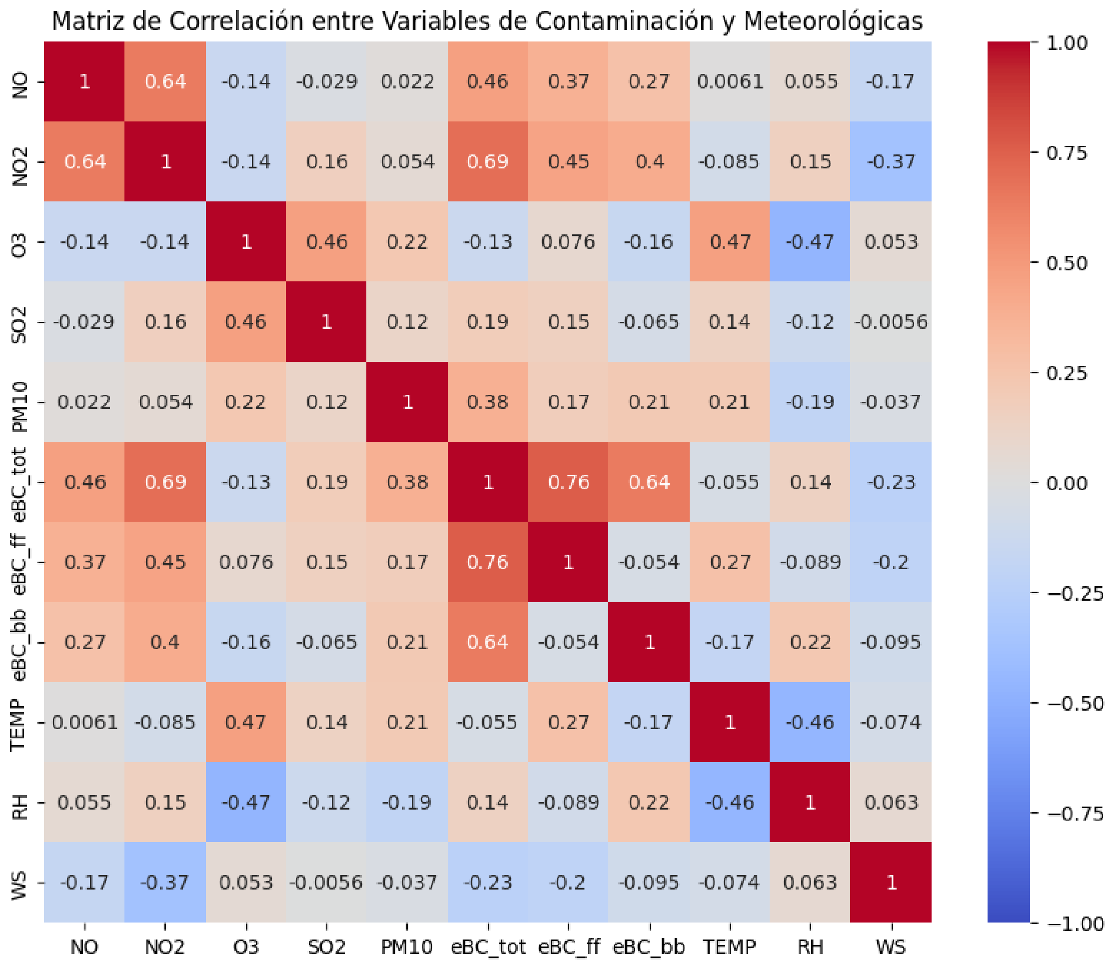


[69]:
```
variables_meteorologicas = ['TEMP', 'RH', 'WS']

# Histograma para Temp, RH, WS
plt.figure(figsize=(15, 5))
for i, variable in enumerate(variables_meteorologicas):
    plt.subplot(1, 3, i+1)
    sns.histplot(trafico_contamina_interseccion[variable], bins=30, kde=True,␣
↪color='lightgreen')
    plt.title(f'Histograma de {variable}')
    plt.xlabel(f'{variable} (Unidades)')
    plt.ylabel('Frecuencia')

plt.tight_layout()
plt.show()
```



1.4 Relaciones entre variables

[70]:
```
# Calcular la matriz de correlación
correlation_matrix = trafico_contamina_interseccion[['NO', 'NO2', 'O3', 'SO2',␣
↪'PM10', 'eBC_tot', 'eBC_ff', 'eBC_bb', 'TEMP', 'RH', 'WS']].corr()

# Mostrar la matriz
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Matriz de Correlación entre Variables de Contaminación y␣
↪Meteorológicas')
plt.show()
```



1.5 Mayores correlaciones

- **eBC_tot-eBC_ff** (0.76) y **eBC_tot-eBC_bb** (0.64):

Las correlaciones más fuertes en los datos de contaminación se observan entre eBC_tot y eBC_ff (0.76), y entre eBC_tot y eBC_bb (0.64). Esto sugiere que eBC_tot, que representa el hollín total, está compuesto principalmente por la fracción atribuible a la quema de combustibles fósiles (eBC_ff), pero también incluye una contribución significativa de la quema de biomasa (eBC_bb). La correlación más alta con eBC_ff indica que el tráfico y otras fuentes de combustibles fósiles son los mayores contribuyentes al black carbon total en la zona estudiada, mientras que la biomasa tiene una menor, pero importante, participación en las concentraciones generales de hollín.

- **NO2** y **eBC_tot** (0.69):
- TODO: **NO** y **NO2** (0.64)

Correlaciones negativas!! - TODO: **RH** y **O3**

1.6 Preprocesamiento

1.6.1 Imputación de datos faltantes con media y moda

[71]:
```
# Imputar valores nulos en variables numéricas con la mediana
numeric_cols = trafico_contamina_interseccion.select_dtypes(include=['float64',␣
↪'int']).columns
trafico_contamina_interseccion[numeric_cols] =␣
↪trafico_contamina_interseccion[numeric_cols].
↪fillna(trafico_contamina_interseccion[numeric_cols].median())

# Imputar valores nulos en variables categóricas con la moda
categorical_cols = trafico_contamina_interseccion.
↪select_dtypes(include=['object']).columns
for col in categorical_cols:
    trafico_contamina_interseccion[col] = trafico_contamina_interseccion[col].
↪fillna(trafico_contamina_interseccion[col].mode()[0])  # Imputar con la moda
```

TODO: Interpolación con métodos numéricos

1.6.2 Eliminación de Duplicados

[72]:
```
trafico_contamina_interseccion.drop_duplicates(inplace=True)
```

[73]:
```
## Esta variables es para dividir el dataset
## Dividir en dos escenarios
## Elaborar dos escenarios: (usar la variable truck_pos)
## 1) Pampaneira
## 2) Capileira
```

1.7 Escalado de datos

[30]:
```
from sklearn.preprocessing import MinMaxScaler

numerical_cols = trafico_contamina_interseccion.select_dtypes(include=('float',␣
↪'int')).columns

# Escalar los datos
scaler = MinMaxScaler()
data_scaled = scaler.
↪fit_transform(trafico_contamina_interseccion[numerical_cols])

# Convertir a DataFrame y combinar
data_scaled_df = pd.DataFrame(data_scaled, columns=numerical_cols)

# Combinar el DataFrame original (sin columnas numéricas) con el DataFrame␣
↪escalado
```

```
result = pd.concat([trafico_contamina_interseccion.drop(columns=numerical_cols).
↪reset_index(drop=True),
                    data_scaled_df.reset_index(drop=True)], axis=1)
```

[31]:
```
df_original = trafico_feb22_ago23
df_interseccion = result

df_interseccion["Date"]
```

[31]:
```
0       2022-11-16 16:00:00
1       2022-11-16 17:00:00
2       2022-11-16 18:00:00
3       2022-11-16 19:00:00
4       2022-11-16 20:00:00
5       2022-11-16 21:00:00
6       2022-11-16 22:00:00
7       2022-11-16 23:00:00
8       2022-11-17 06:00:00
9       2022-11-17 07:00:00
10      2022-11-17 08:00:00
11      2022-11-17 09:00:00
                 …
2165    2023-06-26 12:00:00
2166    2023-06-26 13:00:00
2167    2023-06-26 14:00:00
2168    2023-06-26 15:00:00
2169    2023-06-26 16:00:00
2170    2023-06-26 17:00:00
2171    2023-06-26 18:00:00
2172    2023-06-26 19:00:00
2173    2023-06-26 20:00:00
2174    2023-06-26 21:00:00
2175    2023-06-26 22:00:00
2176    2023-06-27 00:00:00
Name: Date, Length: 2177, dtype: datetime64[ns]
```

[32]:
```
df_interseccion['Date'] = pd.to_datetime(df_interseccion['Date'])

df_interseccion['Date'] = df_interseccion['Date'].dt.tz_localize('UTC')
```

[33]:
```
df_original["date"] = pd.to_datetime(df_original["date"])
```

[34]:
```
df_original["date"] = df_original["date"].dt.tz_convert('UTC')
```

[35]:
```
df_interseccion.rename(columns={'Date': 'date'}, inplace=True)
```

[36]:
```
df_interseccion
```

[36]:
```
                          date        CO  ...  truck_pos_CAP  truck_pos_PAM_2
0     2022-11-16 16:00:00+00:00  0.345133  ...            1.0              0.0
1     2022-11-16 17:00:00+00:00  0.442478  ...            1.0              0.0
2     2022-11-16 18:00:00+00:00  0.256637  ...            1.0              0.0
3     2022-11-16 19:00:00+00:00  0.230088  ...            1.0              0.0
4     2022-11-16 20:00:00+00:00  0.176991  ...            1.0              0.0
5     2022-11-16 21:00:00+00:00  0.150442  ...            1.0              0.0
6     2022-11-16 22:00:00+00:00  0.141593  ...            1.0              0.0
7     2022-11-16 23:00:00+00:00  0.079646  ...            1.0              0.0
8     2022-11-17 06:00:00+00:00  0.026549  ...            1.0              0.0
…                           …       …  …            …              …
2168 2023-06-26 15:00:00+00:00  0.274336  ...            0.0              1.0
2169 2023-06-26 16:00:00+00:00  0.265487  ...            0.0              1.0
2170 2023-06-26 17:00:00+00:00  0.247788  ...            0.0              1.0
2171 2023-06-26 18:00:00+00:00  0.256637  ...            0.0              1.0
2172 2023-06-26 19:00:00+00:00  0.230088  ...            0.0              1.0
2173 2023-06-26 20:00:00+00:00  0.238938  ...            0.0              1.0
2174 2023-06-26 21:00:00+00:00  0.230088  ...            0.0              1.0
2175 2023-06-26 22:00:00+00:00  0.230088  ...            0.0              1.0
2176 2023-06-27 00:00:00+00:00  0.238938  ...            0.0              1.0

[2177 rows x 194 columns]
```

[37]:
```
df_original
```

[37]:
```
                           date  ...  vehicles_BUB_IN_+300_CO2
0      2022-02-04 12:00:00+00:00  ...                         0
1      2022-02-04 13:00:00+00:00  ...                         0
2      2022-02-04 14:00:00+00:00  ...                         0
3      2022-02-04 15:00:00+00:00  ...                         0
4      2022-02-04 16:00:00+00:00  ...                         0
5      2022-02-04 17:00:00+00:00  ...                         0
6      2022-02-04 18:00:00+00:00  ...                         0
7      2022-02-04 19:00:00+00:00  ...                         0
8      2022-02-04 20:00:00+00:00  ...                         0
…                            …  …                       …
10021  2023-08-31 15:00:00+00:00  ...                         0
10022  2023-08-31 16:00:00+00:00  ...                         0
10023  2023-08-31 17:00:00+00:00  ...                         0
10024  2023-08-31 18:00:00+00:00  ...                         0
10025  2023-08-31 19:00:00+00:00  ...                         0
10026  2023-08-31 20:00:00+00:00  ...                         0
10027  2023-08-31 21:00:00+00:00  ...                         0
10028  2023-08-31 22:00:00+00:00  ...                         0
10029  2023-08-31 23:00:00+00:00  ...                         0

[10030 rows x 137 columns]
```

1.8 El conjunto intersección es un subconjunto del original

[38]:
```
set(df_interseccion["date"]).issubset(set(df_original["date"]))
```

[38]:
```
True
```

[39]:
```
df_original[df_original["date"] == df_interseccion["date"][0]]
```

[39]:
```
                           date  …  vehicles_BUB_IN_+300_CO2
4910 2022-11-16 16:00:00+00:00  …                         1

[1 rows x 137 columns]
```

A partir del elemento 4910 se incluye el conjunto intersección en el original

[40]:
```
df_original[df_original["date"] == list(reversed(df_interseccion["date"]))[0]]
```

[40]:
```
                           date  …  vehicles_BUB_IN_+300_CO2
8701 2023-06-27 00:00:00+00:00  …                         0

[1 rows x 137 columns]
```

Hasta el elemento 8701

Cada una de las columnas del dataset original está en el de intersección. Aunque este incluye algunas más (las de contaminación)

[41]:
```
all(column in df_interseccion for column in df_original.columns)
```

[41]:
```
True
```

[87]:
```
columnas_int64 = df_original.select_dtypes(include='int64').columns

# Convertir esas columnas a float64
df_original[columnas_int64] = df_original[columnas_int64].astype('float64')
```

1.9 Fusionar los dos conjuntos

[43]:
```
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)

# Asegúrate de que no haya diferencias por espacios en blanco o tipo de datos
df_original[list(df_original.columns)] = df_original[list(df_original.columns)].
 ↪apply(lambda x: x.str.strip() if x.dtype == "object" else x)
df_interseccion[list(df_original.columns)] = df_interseccion[list(df_original.
 ↪columns)].apply(lambda x: x.str.strip() if x.dtype == "object" else x)
```

[44]:
```
# Identificar las nuevas columnas que quieres rellenar
nuevas_columnas = df_interseccion.columns.difference(df_original.columns)
```

[45]:
```python
# TODO: El merge no rellena los valores

df = pd.merge(df_original, df_interseccion, how="left", on=list(df_original.
 ↪columns))
```

[46]:
```python
fila_inicio = 4910
fila_final = 8701

# Reiniciar los índices para asegurar alineación
df_interseccion.reset_index(drop=True, inplace=True)
df.reset_index(drop=True, inplace=True)

# Comprobar la longitud de las filas seleccionadas
len_interseccion = len(df_interseccion)
len_merged = len(df)

if fila_inicio < len_merged:
    # Asegurarse de que las nuevas columnas existan en df_interseccion
    if all(col in df_interseccion.columns for col in nuevas_columnas):
        # Rellenar df_merged a partir de la fila 4910 con los valores de␣
 ↪df_interseccion
        df.loc[fila_inicio:fila_final, nuevas_columnas] = df_interseccion.
 ↪loc[0, nuevas_columnas].values
    else:
        print("Algunas nuevas columnas no existen en df_interseccion.")
else:
    print("No hay suficientes filas en df_interseccion o df_merged para␣
 ↪realizar la operación.")
```

[47]:
```python
cols = ["date"]
cols.extend(nuevas_columnas)
df.loc[4905:4920, cols]
```

[47]:
```
                          date        CO       NO       NO2        O3  \
4905 2022-11-16 11:00:00+00:00       NaN      NaN       NaN       NaN
4906 2022-11-16 12:00:00+00:00       NaN      NaN       NaN       NaN
4907 2022-11-16 13:00:00+00:00       NaN      NaN       NaN       NaN
4908 2022-11-16 14:00:00+00:00       NaN      NaN       NaN       NaN
4909 2022-11-16 15:00:00+00:00       NaN      NaN       NaN       NaN
4910 2022-11-16 16:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4911 2022-11-16 17:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4912 2022-11-16 18:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4913 2022-11-16 19:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4914 2022-11-16 20:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4915 2022-11-16 21:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4916 2022-11-16 22:00:00+00:00  0.345133  0.02454  0.142857  0.400826
4917 2022-11-16 23:00:00+00:00  0.345133  0.02454  0.142857  0.400826
```

4918	2022-11-17 06:00:00+00:00	0.345133	0.02454	0.142857	0.400826
4919	2022-11-17 07:00:00+00:00	0.345133	0.02454	0.142857	0.400826
4920	2022-11-17 08:00:00+00:00	0.345133	0.02454	0.142857	0.400826

	PM10	PRES	RH	SO2	TEMP	WD	WS \
4905	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4906	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4907	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4908	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4909	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4910	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4911	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4912	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4913	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4914	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4915	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4916	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4917	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4918	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4919	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
4920	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722

	eBC	eBC_bb	eBC_ff	eBC_tot	truck_pos_CAP	truck_pos_PAM_2 \
4905	NaN	NaN	NaN	NaN	NaN	NaN
4906	NaN	NaN	NaN	NaN	NaN	NaN
4907	NaN	NaN	NaN	NaN	NaN	NaN
4908	NaN	NaN	NaN	NaN	NaN	NaN
4909	NaN	NaN	NaN	NaN	NaN	NaN
4910	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4911	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4912	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4913	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4914	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4915	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4916	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4917	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4918	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4919	0.129032	0.045455	0.054054	0.078947	1.0	0.0
4920	0.129032	0.045455	0.054054	0.078947	1.0	0.0

	vehicles_BUB_IN_Etiqueta 0	vehicles_BUB_IN_Etiqueta B \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.25	0.195122

4911	0.25	0.195122
4912	0.25	0.195122
4913	0.25	0.195122
4914	0.25	0.195122
4915	0.25	0.195122
4916	0.25	0.195122
4917	0.25	0.195122
4918	0.25	0.195122
4919	0.25	0.195122
4920	0.25	0.195122

	vehicles_BUB_IN_Etiqueta C	vehicles_BUB_IN_Etiqueta ECO \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.16	0.3
4911	0.16	0.3
4912	0.16	0.3
4913	0.16	0.3
4914	0.16	0.3
4915	0.16	0.3
4916	0.16	0.3
4917	0.16	0.3
4918	0.16	0.3
4919	0.16	0.3
4920	0.16	0.3

	vehicles_BUB_IN_Sin Etiqueta	vehicles_BUB_OUT_Etiqueta 0 \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.1875	0.333333
4911	0.1875	0.333333
4912	0.1875	0.333333
4913	0.1875	0.333333
4914	0.1875	0.333333
4915	0.1875	0.333333
4916	0.1875	0.333333
4917	0.1875	0.333333
4918	0.1875	0.333333
4919	0.1875	0.333333
4920	0.1875	0.333333

	vehicles_BUB_OUT_Etiqueta B	vehicles_BUB_OUT_Etiqueta C	\
4905	NaN	NaN	
4906	NaN	NaN	
4907	NaN	NaN	
4908	NaN	NaN	
4909	NaN	NaN	
4910	0.142857	0.195122	
4911	0.142857	0.195122	
4912	0.142857	0.195122	
4913	0.142857	0.195122	
4914	0.142857	0.195122	
4915	0.142857	0.195122	
4916	0.142857	0.195122	
4917	0.142857	0.195122	
4918	0.142857	0.195122	
4919	0.142857	0.195122	
4920	0.142857	0.195122	

	vehicles_BUB_OUT_Etiqueta ECO	vehicles_BUB_OUT_Sin Etiqueta	\
4905	NaN	NaN	
4906	NaN	NaN	
4907	NaN	NaN	
4908	NaN	NaN	
4909	NaN	NaN	
4910	0.166667	0.290323	
4911	0.166667	0.290323	
4912	0.166667	0.290323	
4913	0.166667	0.290323	
4914	0.166667	0.290323	
4915	0.166667	0.290323	
4916	0.166667	0.290323	
4917	0.166667	0.290323	
4918	0.166667	0.290323	
4919	0.166667	0.290323	
4920	0.166667	0.290323	

	vehicles_CAP_IN_Etiqueta 0	vehicles_CAP_IN_Etiqueta B	\
4905	NaN	NaN	
4906	NaN	NaN	
4907	NaN	NaN	
4908	NaN	NaN	
4909	NaN	NaN	
4910	0.0	0.146341	
4911	0.0	0.146341	
4912	0.0	0.146341	
4913	0.0	0.146341	
4914	0.0	0.146341	

```
4915                       0.0                     0.146341
4916                       0.0                     0.146341
4917                       0.0                     0.146341
4918                       0.0                     0.146341
4919                       0.0                     0.146341
4920                       0.0                     0.146341

      vehicles_CAP_IN_Etiqueta C  vehicles_CAP_IN_Etiqueta ECO  \
4905                         NaN                           NaN
4906                         NaN                           NaN
4907                         NaN                           NaN
4908                         NaN                           NaN
4909                         NaN                           NaN
4910                    0.181818                      0.076923
4911                    0.181818                      0.076923
4912                    0.181818                      0.076923
4913                    0.181818                      0.076923
4914                    0.181818                      0.076923
4915                    0.181818                      0.076923
4916                    0.181818                      0.076923
4917                    0.181818                      0.076923
4918                    0.181818                      0.076923
4919                    0.181818                      0.076923
4920                    0.181818                      0.076923

      vehicles_CAP_IN_Sin Etiqueta  vehicles_CAP_OUT_Etiqueta 0  \
4905                           NaN                          NaN
4906                           NaN                          NaN
4907                           NaN                          NaN
4908                           NaN                          NaN
4909                           NaN                          NaN
4910                      0.178571                     0.333333
4911                      0.178571                     0.333333
4912                      0.178571                     0.333333
4913                      0.178571                     0.333333
4914                      0.178571                     0.333333
4915                      0.178571                     0.333333
4916                      0.178571                     0.333333
4917                      0.178571                     0.333333
4918                      0.178571                     0.333333
4919                      0.178571                     0.333333
4920                      0.178571                     0.333333

      vehicles_CAP_OUT_Etiqueta B  vehicles_CAP_OUT_Etiqueta C  \
4905                          NaN                          NaN
4906                          NaN                          NaN
4907                          NaN                          NaN
```

4908	NaN	NaN
4909	NaN	NaN
4910	0.135135	0.170213
4911	0.135135	0.170213
4912	0.135135	0.170213
4913	0.135135	0.170213
4914	0.135135	0.170213
4915	0.135135	0.170213
4916	0.135135	0.170213
4917	0.135135	0.170213
4918	0.135135	0.170213
4919	0.135135	0.170213
4920	0.135135	0.170213

	vehicles_CAP_OUT_Etiqueta ECO	vehicles_CAP_OUT_Sin Etiqueta \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.0	0.375
4911	0.0	0.375
4912	0.0	0.375
4913	0.0	0.375
4914	0.0	0.375
4915	0.0	0.375
4916	0.0	0.375
4917	0.0	0.375
4918	0.0	0.375
4919	0.0	0.375
4920	0.0	0.375

	vehicles_PAM_1_IN_Etiqueta 0	vehicles_PAM_1_IN_Etiqueta B \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.0	0.125
4911	0.0	0.125
4912	0.0	0.125
4913	0.0	0.125
4914	0.0	0.125
4915	0.0	0.125
4916	0.0	0.125
4917	0.0	0.125
4918	0.0	0.125

4919	0.0	0.125
4920	0.0	0.125

	vehicles_PAM_1_IN_Etiqueta C	vehicles_PAM_1_IN_Etiqueta ECO \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.101266	0.153846
4911	0.101266	0.153846
4912	0.101266	0.153846
4913	0.101266	0.153846
4914	0.101266	0.153846
4915	0.101266	0.153846
4916	0.101266	0.153846
4917	0.101266	0.153846
4918	0.101266	0.153846
4919	0.101266	0.153846
4920	0.101266	0.153846

	vehicles_PAM_1_IN_Sin Etiqueta	vehicles_PAM_1_OUT_Etiqueta 0 \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.225	0.0
4911	0.225	0.0
4912	0.225	0.0
4913	0.225	0.0
4914	0.225	0.0
4915	0.225	0.0
4916	0.225	0.0
4917	0.225	0.0
4918	0.225	0.0
4919	0.225	0.0
4920	0.225	0.0

	vehicles_PAM_1_OUT_Etiqueta B	vehicles_PAM_1_OUT_Etiqueta C \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.363636	0.384615
4911	0.363636	0.384615

4912	0.363636	0.384615
4913	0.363636	0.384615
4914	0.363636	0.384615
4915	0.363636	0.384615
4916	0.363636	0.384615
4917	0.363636	0.384615
4918	0.363636	0.384615
4919	0.363636	0.384615
4920	0.363636	0.384615

	vehicles_PAM_1_OUT_Etiqueta ECO	vehicles_PAM_1_OUT_Sin Etiqueta \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.2	0.405405
4911	0.2	0.405405
4912	0.2	0.405405
4913	0.2	0.405405
4914	0.2	0.405405
4915	0.2	0.405405
4916	0.2	0.405405
4917	0.2	0.405405
4918	0.2	0.405405
4919	0.2	0.405405
4920	0.2	0.405405

	vehicles_PAM_2_IN_Etiqueta 0	vehicles_PAM_2_IN_Etiqueta B \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.0	0.27907
4911	0.0	0.27907
4912	0.0	0.27907
4913	0.0	0.27907
4914	0.0	0.27907
4915	0.0	0.27907
4916	0.0	0.27907
4917	0.0	0.27907
4918	0.0	0.27907
4919	0.0	0.27907
4920	0.0	0.27907

	vehicles_PAM_2_IN_Etiqueta C	vehicles_PAM_2_IN_Etiqueta ECO \

4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.301887	0.125
4911	0.301887	0.125
4912	0.301887	0.125
4913	0.301887	0.125
4914	0.301887	0.125
4915	0.301887	0.125
4916	0.301887	0.125
4917	0.301887	0.125
4918	0.301887	0.125
4919	0.301887	0.125
4920	0.301887	0.125

	vehicles_PAM_2_IN_Sin Etiqueta	vehicles_PAM_2_OUT_Etiqueta 0 \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.548387	0.0
4911	0.548387	0.0
4912	0.548387	0.0
4913	0.548387	0.0
4914	0.548387	0.0
4915	0.548387	0.0
4916	0.548387	0.0
4917	0.548387	0.0
4918	0.548387	0.0
4919	0.548387	0.0
4920	0.548387	0.0

	vehicles_PAM_2_OUT_Etiqueta B	vehicles_PAM_2_OUT_Etiqueta C \
4905	NaN	NaN
4906	NaN	NaN
4907	NaN	NaN
4908	NaN	NaN
4909	NaN	NaN
4910	0.21875	0.208333
4911	0.21875	0.208333
4912	0.21875	0.208333
4913	0.21875	0.208333
4914	0.21875	0.208333
4915	0.21875	0.208333

```
4916                      0.21875                       0.208333
4917                      0.21875                       0.208333
4918                      0.21875                       0.208333
4919                      0.21875                       0.208333
4920                      0.21875                       0.208333

      vehicles_PAM_2_OUT_Etiqueta ECO  vehicles_PAM_2_OUT_Sin Etiqueta
4905                              NaN                              NaN
4906                              NaN                              NaN
4907                              NaN                              NaN
4908                              NaN                              NaN
4909                              NaN                              NaN
4910                            0.625                              0.3
4911                            0.625                              0.3
4912                            0.625                              0.3
4913                            0.625                              0.3
4914                            0.625                              0.3
4915                            0.625                              0.3
4916                            0.625                              0.3
4917                            0.625                              0.3
4918                            0.625                              0.3
4919                            0.625                              0.3
4920                            0.625                              0.3
```

[48]:
```
df.loc[8695:8710,cols]
```

[48]:
```
                          date        CO       NO       NO2        O3  \
8695 2023-06-26 17:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8696 2023-06-26 18:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8697 2023-06-26 19:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8698 2023-06-26 20:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8699 2023-06-26 21:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8700 2023-06-26 22:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8701 2023-06-27 00:00:00+00:00  0.345133  0.02454  0.142857  0.400826
8702 2023-06-27 02:00:00+00:00       NaN      NaN       NaN       NaN
8703 2023-06-27 06:00:00+00:00       NaN      NaN       NaN       NaN
8704 2023-06-27 07:00:00+00:00       NaN      NaN       NaN       NaN
8705 2023-06-27 08:00:00+00:00       NaN      NaN       NaN       NaN
8706 2023-06-27 09:00:00+00:00       NaN      NaN       NaN       NaN
8707 2023-06-27 10:00:00+00:00       NaN      NaN       NaN       NaN
8708 2023-06-27 11:00:00+00:00       NaN      NaN       NaN       NaN
8709 2023-06-27 12:00:00+00:00       NaN      NaN       NaN       NaN
8710 2023-06-27 13:00:00+00:00       NaN      NaN       NaN       NaN

          PM10      PRES        RH       SO2      TEMP        WD        WS  \
8695  0.068464  0.179894  0.581114  0.413978  0.397059  0.395132  0.369722
8696  0.068464  0.179894  0.581114  0.413978  0.397059  0.395132  0.369722
```

8697	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
8698	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
8699	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
8700	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
8701	0.068464	0.179894	0.581114	0.413978	0.397059	0.395132	0.369722
8702	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8703	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8704	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8705	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8706	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8707	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8708	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8709	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8710	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	eBC	eBC_bb	eBC_ff	eBC_tot	truck_pos_CAP	truck_pos_PAM_2 \
8695	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8696	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8697	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8698	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8699	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8700	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8701	0.129032	0.045455	0.054054	0.078947	1.0	0.0
8702	NaN	NaN	NaN	NaN	NaN	NaN
8703	NaN	NaN	NaN	NaN	NaN	NaN
8704	NaN	NaN	NaN	NaN	NaN	NaN
8705	NaN	NaN	NaN	NaN	NaN	NaN
8706	NaN	NaN	NaN	NaN	NaN	NaN
8707	NaN	NaN	NaN	NaN	NaN	NaN
8708	NaN	NaN	NaN	NaN	NaN	NaN
8709	NaN	NaN	NaN	NaN	NaN	NaN
8710	NaN	NaN	NaN	NaN	NaN	NaN

	vehicles_BUB_IN_Etiqueta 0	vehicles_BUB_IN_Etiqueta B \
8695	0.25	0.195122
8696	0.25	0.195122
8697	0.25	0.195122
8698	0.25	0.195122
8699	0.25	0.195122
8700	0.25	0.195122
8701	0.25	0.195122
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN

```
8708                          NaN                         NaN
8709                          NaN                         NaN
8710                          NaN                         NaN

      vehicles_BUB_IN_Etiqueta C  vehicles_BUB_IN_Etiqueta ECO  \
8695                        0.16                           0.3
8696                        0.16                           0.3
8697                        0.16                           0.3
8698                        0.16                           0.3
8699                        0.16                           0.3
8700                        0.16                           0.3
8701                        0.16                           0.3
8702                         NaN                           NaN
8703                         NaN                           NaN
8704                         NaN                           NaN
8705                         NaN                           NaN
8706                         NaN                           NaN
8707                         NaN                           NaN
8708                         NaN                           NaN
8709                         NaN                           NaN
8710                         NaN                           NaN

      vehicles_BUB_IN_Sin Etiqueta  vehicles_BUB_OUT_Etiqueta 0  \
8695                        0.1875                     0.333333
8696                        0.1875                     0.333333
8697                        0.1875                     0.333333
8698                        0.1875                     0.333333
8699                        0.1875                     0.333333
8700                        0.1875                     0.333333
8701                        0.1875                     0.333333
8702                           NaN                          NaN
8703                           NaN                          NaN
8704                           NaN                          NaN
8705                           NaN                          NaN
8706                           NaN                          NaN
8707                           NaN                          NaN
8708                           NaN                          NaN
8709                           NaN                          NaN
8710                           NaN                          NaN

      vehicles_BUB_OUT_Etiqueta B  vehicles_BUB_OUT_Etiqueta C  \
8695                     0.142857                     0.195122
8696                     0.142857                     0.195122
8697                     0.142857                     0.195122
8698                     0.142857                     0.195122
8699                     0.142857                     0.195122
8700                     0.142857                     0.195122
```

8701	0.142857	0.195122
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_BUB_OUT_Etiqueta ECO	vehicles_BUB_OUT_Sin Etiqueta \
8695	0.166667	0.290323
8696	0.166667	0.290323
8697	0.166667	0.290323
8698	0.166667	0.290323
8699	0.166667	0.290323
8700	0.166667	0.290323
8701	0.166667	0.290323
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_CAP_IN_Etiqueta 0	vehicles_CAP_IN_Etiqueta B \
8695	0.0	0.146341
8696	0.0	0.146341
8697	0.0	0.146341
8698	0.0	0.146341
8699	0.0	0.146341
8700	0.0	0.146341
8701	0.0	0.146341
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_CAP_IN_Etiqueta C	vehicles_CAP_IN_Etiqueta ECO	\
8695	0.181818	0.076923	
8696	0.181818	0.076923	
8697	0.181818	0.076923	
8698	0.181818	0.076923	
8699	0.181818	0.076923	
8700	0.181818	0.076923	
8701	0.181818	0.076923	
8702	NaN	NaN	
8703	NaN	NaN	
8704	NaN	NaN	
8705	NaN	NaN	
8706	NaN	NaN	
8707	NaN	NaN	
8708	NaN	NaN	
8709	NaN	NaN	
8710	NaN	NaN	

	vehicles_CAP_IN_Sin Etiqueta	vehicles_CAP_OUT_Etiqueta 0	\
8695	0.178571	0.333333	
8696	0.178571	0.333333	
8697	0.178571	0.333333	
8698	0.178571	0.333333	
8699	0.178571	0.333333	
8700	0.178571	0.333333	
8701	0.178571	0.333333	
8702	NaN	NaN	
8703	NaN	NaN	
8704	NaN	NaN	
8705	NaN	NaN	
8706	NaN	NaN	
8707	NaN	NaN	
8708	NaN	NaN	
8709	NaN	NaN	
8710	NaN	NaN	

	vehicles_CAP_OUT_Etiqueta B	vehicles_CAP_OUT_Etiqueta C	\
8695	0.135135	0.170213	
8696	0.135135	0.170213	
8697	0.135135	0.170213	
8698	0.135135	0.170213	
8699	0.135135	0.170213	
8700	0.135135	0.170213	
8701	0.135135	0.170213	
8702	NaN	NaN	
8703	NaN	NaN	
8704	NaN	NaN	

8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_CAP_OUT_Etiqueta ECO	vehicles_CAP_OUT_Sin Etiqueta \
8695	0.0	0.375
8696	0.0	0.375
8697	0.0	0.375
8698	0.0	0.375
8699	0.0	0.375
8700	0.0	0.375
8701	0.0	0.375
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_1_IN_Etiqueta 0	vehicles_PAM_1_IN_Etiqueta B \
8695	0.0	0.125
8696	0.0	0.125
8697	0.0	0.125
8698	0.0	0.125
8699	0.0	0.125
8700	0.0	0.125
8701	0.0	0.125
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_1_IN_Etiqueta C	vehicles_PAM_1_IN_Etiqueta ECO \
8695	0.101266	0.153846
8696	0.101266	0.153846
8697	0.101266	0.153846

8698	0.101266	0.153846
8699	0.101266	0.153846
8700	0.101266	0.153846
8701	0.101266	0.153846
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_1_IN_Sin Etiqueta	vehicles_PAM_1_OUT_Etiqueta 0	\
8695	0.225	0.0	
8696	0.225	0.0	
8697	0.225	0.0	
8698	0.225	0.0	
8699	0.225	0.0	
8700	0.225	0.0	
8701	0.225	0.0	
8702	NaN	NaN	
8703	NaN	NaN	
8704	NaN	NaN	
8705	NaN	NaN	
8706	NaN	NaN	
8707	NaN	NaN	
8708	NaN	NaN	
8709	NaN	NaN	
8710	NaN	NaN	

	vehicles_PAM_1_OUT_Etiqueta B	vehicles_PAM_1_OUT_Etiqueta C	\
8695	0.363636	0.384615	
8696	0.363636	0.384615	
8697	0.363636	0.384615	
8698	0.363636	0.384615	
8699	0.363636	0.384615	
8700	0.363636	0.384615	
8701	0.363636	0.384615	
8702	NaN	NaN	
8703	NaN	NaN	
8704	NaN	NaN	
8705	NaN	NaN	
8706	NaN	NaN	
8707	NaN	NaN	
8708	NaN	NaN	

```
8709                            NaN                           NaN
8710                            NaN                           NaN

      vehicles_PAM_1_OUT_Etiqueta ECO  vehicles_PAM_1_OUT_Sin Etiqueta  \
8695                              0.2                         0.405405
8696                              0.2                         0.405405
8697                              0.2                         0.405405
8698                              0.2                         0.405405
8699                              0.2                         0.405405
8700                              0.2                         0.405405
8701                              0.2                         0.405405
8702                              NaN                              NaN
8703                              NaN                              NaN
8704                              NaN                              NaN
8705                              NaN                              NaN
8706                              NaN                              NaN
8707                              NaN                              NaN
8708                              NaN                              NaN
8709                              NaN                              NaN
8710                              NaN                              NaN

      vehicles_PAM_2_IN_Etiqueta 0  vehicles_PAM_2_IN_Etiqueta B  \
8695                           0.0                       0.27907
8696                           0.0                       0.27907
8697                           0.0                       0.27907
8698                           0.0                       0.27907
8699                           0.0                       0.27907
8700                           0.0                       0.27907
8701                           0.0                       0.27907
8702                           NaN                           NaN
8703                           NaN                           NaN
8704                           NaN                           NaN
8705                           NaN                           NaN
8706                           NaN                           NaN
8707                           NaN                           NaN
8708                           NaN                           NaN
8709                           NaN                           NaN
8710                           NaN                           NaN

      vehicles_PAM_2_IN_Etiqueta C  vehicles_PAM_2_IN_Etiqueta ECO  \
8695                      0.301887                           0.125
8696                      0.301887                           0.125
8697                      0.301887                           0.125
8698                      0.301887                           0.125
8699                      0.301887                           0.125
8700                      0.301887                           0.125
8701                      0.301887                           0.125
```

8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_2_IN_Sin Etiqueta	vehicles_PAM_2_OUT_Etiqueta 0 \
8695	0.548387	0.0
8696	0.548387	0.0
8697	0.548387	0.0
8698	0.548387	0.0
8699	0.548387	0.0
8700	0.548387	0.0
8701	0.548387	0.0
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_2_OUT_Etiqueta B	vehicles_PAM_2_OUT_Etiqueta C \
8695	0.21875	0.208333
8696	0.21875	0.208333
8697	0.21875	0.208333
8698	0.21875	0.208333
8699	0.21875	0.208333
8700	0.21875	0.208333
8701	0.21875	0.208333
8702	NaN	NaN
8703	NaN	NaN
8704	NaN	NaN
8705	NaN	NaN
8706	NaN	NaN
8707	NaN	NaN
8708	NaN	NaN
8709	NaN	NaN
8710	NaN	NaN

	vehicles_PAM_2_OUT_Etiqueta ECO	vehicles_PAM_2_OUT_Sin Etiqueta

```
8695                               0.625                                   0.3
8696                               0.625                                   0.3
8697                               0.625                                   0.3
8698                               0.625                                   0.3
8699                               0.625                                   0.3
8700                               0.625                                   0.3
8701                               0.625                                   0.3
8702                                 NaN                                   NaN
8703                                 NaN                                   NaN
8704                                 NaN                                   NaN
8705                                 NaN                                   NaN
8706                                 NaN                                   NaN
8707                                 NaN                                   NaN
8708                                 NaN                                   NaN
8709                                 NaN                                   NaN
8710                                 NaN                                   NaN
```

1.10 Prioritario:

Entrenar el transformer con el subconjunto (funciona bien) Hacer el merge de datos

- Hacer una matriz una de correlacion de las variables de contaminación y el tráfico.
- Debería una seleccionar una variable NO, NO2 y eBM_tot (Black carbon)
- Antes de hacer el merge, transformer para aumentar la muestra de los datos de contaminación. (n+200)
- Entrenar un modelo de regresión o clasificación para las variables elegidas.
- Métricas de clasificación o regresión (entropía -> fundamentos matemáticos)

[51]:
```python
# Calcular la matriz de correlac

correlation_matrix = df.corr()
# Mostrar la matriz
plt.figure(figsize=(100, 80))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Matriz de Correlación entre Variables de Contaminación y entrada y␣
 ↪salida de vehiculos')
plt.show()
```